

UPSC Prelims

Environment & Ecology

Class No. 7

Important Species

- **Context**
- Eight wild African Cheetahs (5 female and 3 male) from Namibia, were recently introduced in Kuno National Park (KNP), Madhya Pradesh under Project Cheetah.
- Another 12 cheetahs were brought in from South Africa recently
- Cheetah was declared extinct from India in 1952 becoming only large wild mammalian species that went extinct since independence (1947)
- **African Cheetah**
- Both a flagship species (selected to act as an ambassador, icon or symbol for a defined habitat, issue, campaign or environmental cause) and an umbrella species (its conservation ensures conservation of other species of wild animals and forests).

Cheetah

	African Cheetah	Asiatic Cheetah
Pictorial Representation		
IUCN Status	Vulnerable	Critically Endangered
CITES Status	Appendix I	Appendix I
Distribution	Africa (Northwest Africa, East Africa, and Southern Africa)	Only few left in Iran
Physical Characteristics	Bigger in size as compared to Asiatic Cheetah, slightly bigger build and sturdy legs and neck.	Slightly smaller and slender than the African Cheetah. Their neck is much smaller and slender. Also, their legs are slender.
Food Intake	Diverse diet due to bigger habitat	Limited source especially medium sized prey like Chinkara, Gazelle etc.

Cheetah

- Characteristics:
- World's fastest land mammal, capable of running at 80 to 128 km/h.
- Diurnal i.e., they hunt during the day.
- Unlike other big cats (lions, tigers, etc.) don't roar.
- Gestation period- 93 days.



- **Live in three main social groups:** females and their cubs, male coalitions, and solitary males.
- Females lead a nomadic life searching for prey in large home ranges, males are more sedentary and instead establish much smaller territories.
- Average female home ranges have been estimated to be about 750 sq km.
- Among large carnivores, conflicts with human interests are lowest for cheetahs, as they are not a threat to humans and usually do not attack large livestock.

- **Project Cheetah**
- World's first inter-continental large wild carnivore translocation project.
- Part of the centrally Sponsored Scheme- 'Project Tiger' of India.
- **Funding:** Funds from Project Tiger and Compensatory Afforestation Fund Management & Planning Authority (CAMPA)
- **Nodal Agency:** National Tiger Conservation Authority (NTCA) authorized for funding, supervision and hand holding of the project.

- Wildlife Institute of India (WII), national and international carnivore/cheetah experts/agencies for technical and knowledge support.
- Significance of Cheetah Relocation:
- Establish viable cheetah metapopulation in India
- Cheetahs can help restore open forest and grassland ecosystems in India
- Resource mobilization, contributing to global conservation efforts
- Enhance local community livelihoods, contribute to climate change mitigation goals etc

- **Why Kuno National Park?**
- No human settlements: There has been a complete relocation of roughly 24 villages and their
- Domesticated livestock from inside the park years ago.
- **Savannah habitat:** The village sites and agricultural fields have now been taken over by grasses.
- **Range:** The region is very close to the Sal forests of Chhattisgarh, the historical range of the cheetah.
- **Scope for coexistence:** Kuno offers the prospect of housing all four large felines in India — tiger, lion, leopard and cheetah.
- Kuno was originally proposed to provide a second home for lions.
- All cheetahs to be vaccinated and fitted with satellite radio collars.
- Action plan for cheetah prepared on the basis of IUCN guidelines.

Cheetah



Terms	Meaning
Soft release	Cheetahs placed in a quarantine period for a month, in an electronically-fenced area of 500 hectares, before being released into the wild.
Cheetah mitras	Dedicated team of volunteers, trained by the forest department trained to create awareness among the villagers about protecting cheetahs.
Chintu Cheetah	Local mascot to sensitize locals
Cheetah Task Force:	Constituted by MoEFCC, for 2 years, its function would include:
	Reviewing, progressing and monitoring the health status of Cheetah
	Suggesting and advising on development of tourism infrastructure
	Regularly interacting with Cheetah mitras and local communities for their awareness raising

About Kuno National Park (KNP)

- **Geographical location and other Important features of KNP:**

- **Location:** Lies in North-West of Madhya Pradesh close to state border with Rajasthan in Vindhyan Hills of Central India.
- On its south-eastern side of the boundary, it forms a contiguous forest landscape with patchy connectivity to **Panna Tiger reserve (TR)** through the **Shivpuri forest area**.
- **Ranthambhore NP and Kailadevi WLS** (both part of the Ranthambhore TR) are connected with it through good forest patches towards the north- western boundary from across the river Chambal.
- Only wildlife site in the country with a complete incentivized voluntary relocation of villages from inside the park.

- **River:** Perennial Kuno river

- **Forest Type:** Northern tropical dry deciduous forest

- **Flora and Fauna:**

- **Trees like Kardhai, Salai, Khair etc.**
- **Herbivores** include- sambar, nilgai, wild pig (*Sus scrofa*), chinkara, chousingha, blackbuck etc.
- **Carnivores** include leopard, sloth bear, striped hyaena, gray wolf, golden jackal, Indian fox etc.



Asiatic Lion

- The lion is one of five pantherine cats inhabiting India, along with the Bengal tiger, Indian leopard, snow leopard and clouded leopard.
- Their population is limited to only five protected areas in Gujarat – Gir National Park, Gir Sanctuary, Pania Sanctuary, Mitiyala Sanctuary and Girnar Sanctuary.
- MoEFCC launched the “Asiatic Lion Conservation Project” with an aim to protect and conserve the world’s last ranging free population of Asiatic Lion
- Threat: Vulnerable to disease, genetic inbreeding, disaster, potential poaching and accidental lion deaths due to human causes.
- Recently 23 Lions died in short period of 20 days, due to Canine Distemper Virus (CDV) and tickbore Babesiosis

State animal- Gujarat



Asiatic Lion



- **Characteristics:**
- Second-largest cats in the world, after tigers.
- Social cats which live in groups called prides
- Male lions have a distinctive thick mane of hair around their heads that females lack
- Recently prepared document : Lion @47, vision for amrit kaal.

Asiatic Lion



Difference between Asiatic and African Lions		
Criteria	Asiatic	African
Size	Larger	Smaller
Mane	Relatively dense, Lighter mane	Relatively short, Darker, sparse mane
Skin Fold	Absent	Longitudinal fold of skin that runs along the belly
Pride Size	Larger	Smaller

Asiatic Lion



Greater One-Horned Rhino

- **Habitat**
- Alluvial Terai-Duar savanna grasslands and riverine forest.
- Found commonly in Nepal, Bhutan, Pakistan and India.
- India home to over 85% of the population.
- Mainly found in the seven Protected Areas viz. –
- Kaziranga NP, Pobitara WLS, Orang NP, Manas NP in Assam, Jaldapara NP and Gorumara NP in West Bengal and Dudhwa NP in Uttar Pradesh.
- Assam has 71% of the world's population (2652 as per 2018 census) with Kaziranga National Park harbouring the highest number.
- State animal- Assam



Greater One-Horned Rhino



- **Characteristics:**
- Asia's largest Rhino species and 4th largest land animal.
- Excellent swimmer and can run at a speed of 55 km/hr for short periods.
- Solitary animal, though several may occupy the same patch of grassland or water source.
- Has incredible sense of hearing and smell, but eyesight is relatively poor.
- Gestation period is about 16 months.
- **Threats:** Hunted for their horn (an ingredient in traditional Asian medicines), destruction of their preferred habitat

Greater One-Horned Rhino



Greater One-Horned Rhino



- **Rhino Species Across the World**
- **White Rhino and Black Rhino:** found in Africa. Black Rhino is the smaller of the two.
- **Javan Rhino:** only a few survive in Java and Vietnam.
- **Sumatran Rhino:** Sumatran rhino is the smallest species of rhinoceros. 30 to 80 Sumatran rhinos remain in the world, mainly on the Indonesian island of Sumatra.
- **Great One-Horned Rhino:** Only the Great One-Horned Rhino is found in India. It is the largest of the rhino species (second in size only to the Asian elephant).
- African and Sumatran rhinoceros have two horns, while the others (Indian and Javan) have a single horn.
- The Indian rhino's horn is present in both males and females.

Greater One-Horned Rhino



- **India Rhino Vision (IRV) 2020**
- Assam adopted the ambitious Indian Rhino Vision (IRV) 2020 program in partnership with WWF India and the International Rhino Foundation.
- Vision was to build a 3000 wild population of Greater One Horned Rhinos by 2020, spread across 7 protected areas in Assam,
- Kaziranga has concentrated population of rhinos which called for spreading of its population
- The Indian Rhino Vision 2020 ended with the release of two rhinos in Assam's Manas NP transported from Pobitra WLS.
- IRV 2020 is believed to have achieved its target of attaining a population of 3000 rhinos in Assam

Snow Leopard



- **Characteristics:**

- Termed as Ghost of Mountain due to their camouflage.
- Patterned with black spots called rosettes unique to every snow leopard.
- A flagship species of high-altitude Himalayas in India.

- **Conservation Measures:**

- Declared as State animal of Himachal Pradesh and Ladakh.
- Snow Leopard Population Assessment in India (SPAI)
- Project Snow Leopard (2009), a centrally supported program for conservation.
- SECURE Himalaya project (2017) by Government of India and UNDP with support of Global Environment Facility.

Snow Leopard



Snow Leopard



- India has ratified the Global Snow Leopard and Ecosystem Protection Program (GSLEP).
- **Global Snow Leopard and Ecosystem Program (GSLESP)**
- In 2013, the scope of Global Tiger Initiative was broadened to include Snow Leopards. Member countries adopted Bishkek Declaration that aims members to work together to identify and secure at least 20 snow leopard landscapes across the cat's range by 2020 also called "Secure 20 by 2020."

Snow Leopard



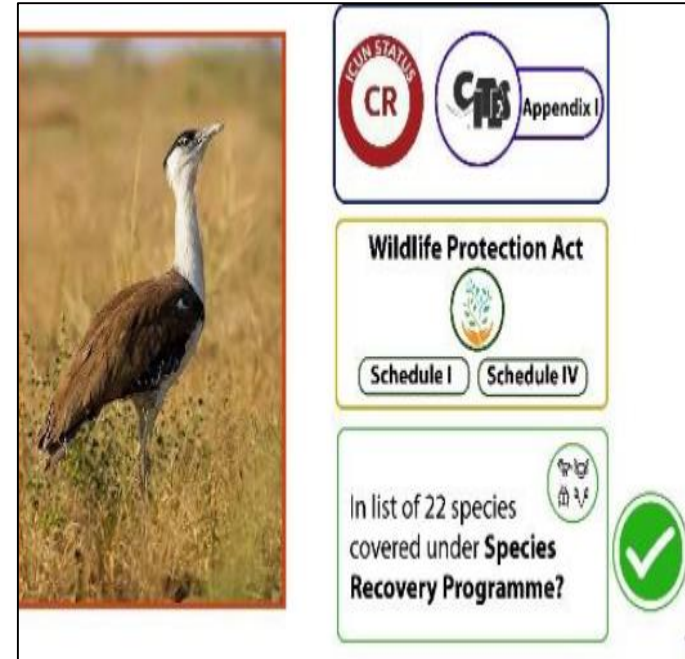
- **Habitat:**
- Mountainous landscape of Central and Southern Asia with 12 snow leopard range countries being- Afghanistan, Bhutan, China, India, Kazakhstan, Kyrgyz Republic, Mongolia, Nepal, Pakistan, Russia, Tajikistan, and Uzbekistan.
- China has world's largest snow leopard population.
- In India, they are found in:
 - Western Himalayas, including the Union Territories of Jammu and Kashmir and Ladakh, Himachal Pradesh, Uttarakhand
 - Sikkim and Arunachal Pradesh in the eastern Himalayas.

Snow Leopard



Great Indian Bustard

- **Context**
- Supreme Court, in April, 2021 had ordered the power companies in Rajasthan and Gujarat to make the high-tension power lines underground so that GIB do not get caught in the web.
- **Characteristics:**
- Called ‘Son Chiriya’ in Madhya Pradesh, ‘Godawan’ in Rajasthan and ‘Maldhok’ in Maharashtra.
- Looks like an ostrich.
- Despite weighing heavy, can easily fly.
- Bustard Species Found in India: Great Indian Bustard, the Lesser Florican and the Bengal Florican.

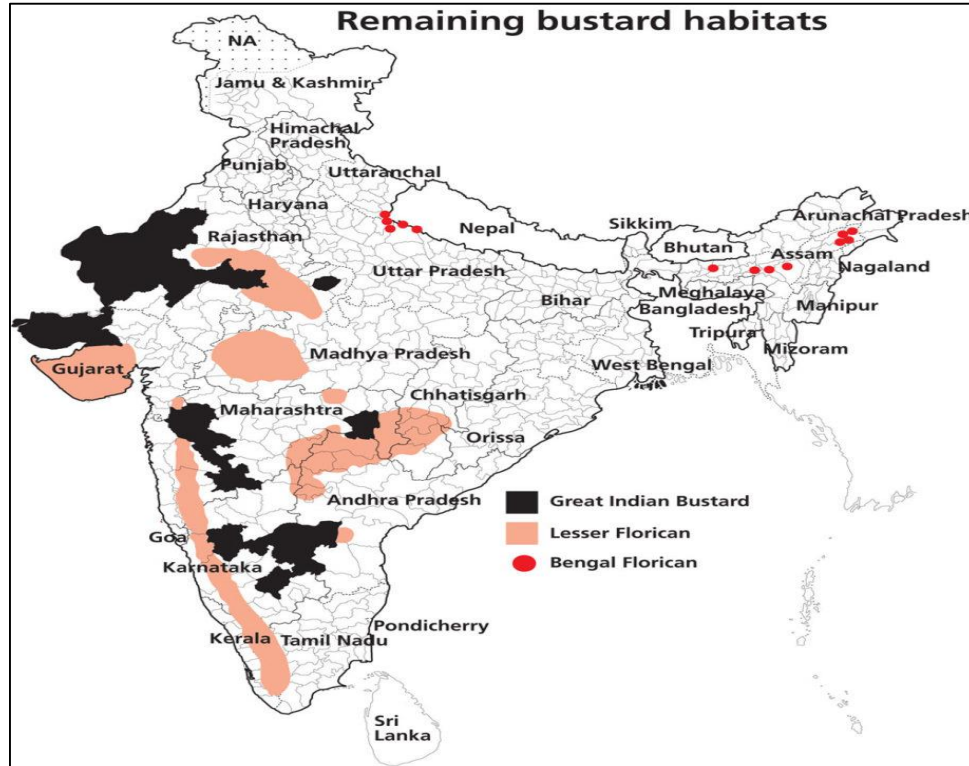


Great Indian Bustard



- **Habitat:**
- Arid and semi-arid grasslands, open country with thorn scrub, tall grass interspersed with cultivation.
- Avoids irrigated areas.
- Endemic to Indian Sub-continent, Rajasthan has the highest population.
- Important Sites: Desert National Park Sanctuary (Rajasthan), Naliya (Gujarat), Warora (Maharashtra) and Bellary (Karnataka)
- **Threats to GIB:**
- Hunting, habitat erosion, 'greening' projects that transform arid grasslands to wooded areas, predators preying on the eggs, power transmission lines etc

Great Indian Bustard



Please take note of distribution of lesser florican.

Gharial

- Derives its name from ghara, an Indian word for pot because of a bulbous knob (narial excrescence) present at the end of their snout.

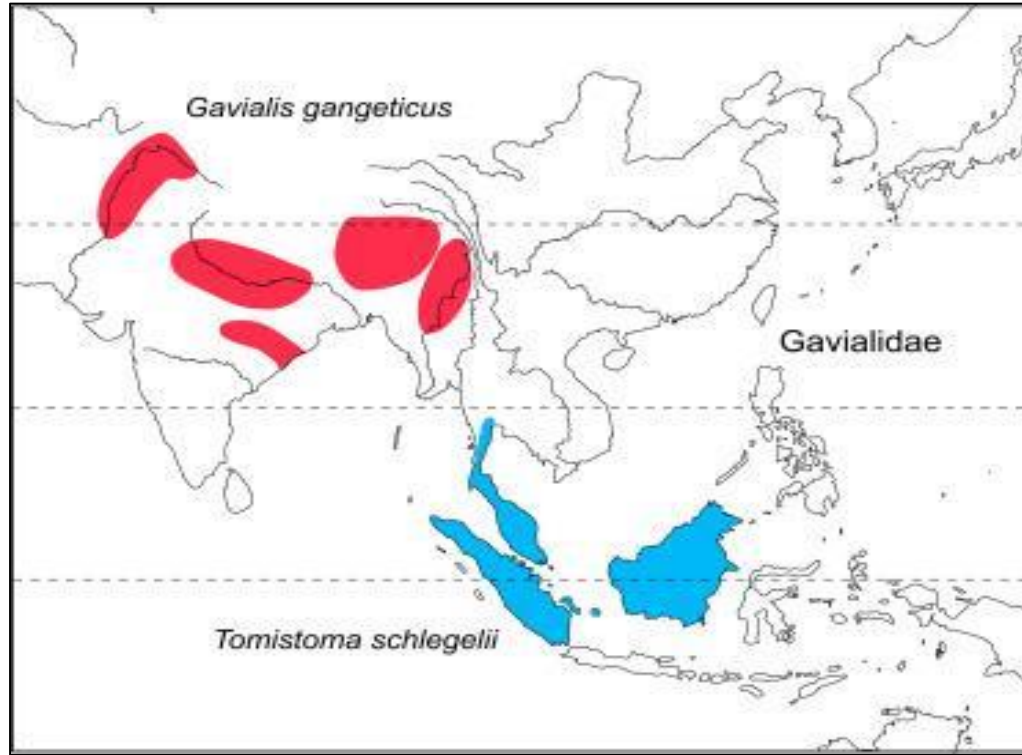
A	<i>Common Name</i> Gharial
Aa	<i>Scientific Name</i> <i>Gavialis gangeticus</i>
	<i>Length</i> 3-6 meter (Male), 2.5-4 meter (Female)
	<i>Weight</i> 150-250 Kg
	<i>Population</i> Approximately 800
	<i>Status</i> Listed in Schedule I of Wildlife (Protection) Act, 1972 and as Critically Endangered on IUCN Red List

Gharial

- **Habitat distribution:**
- Ramganga, River Girwa, River Gandaki, River Chambal, River Ganges, Son River, Mahanadi River in Odisha. Nepal: Rapti-Narayani River
- **Conservation Efforts:**
- Breeding Centres of Kukrail
- Gharial Rehabilitation Centre in Lucknow, Uttar Pradesh
- National Chambal Sanctuary (Gharial Eco Park, Madhya Pradesh).



Gharial



- **Threats**
- Dam, barrages, and water abstraction adversely affects gharial by turning suitable river habitats into marginal/ unsuitable lakes, and by altering the quantity and quality of water available to downstream river sections.
- Gharial, with its long, toothy rostrum is particularly vulnerable to entanglement in fishing nets, where it is frequently trapped underwater and drowns.

Mugger

- The mugger (marsh/Indian/Broad Snouted) crocodile (VU) is a freshwater species.
- Habitat: freshwater lakes, rivers and marshes, reservoirs, and slow-moving, shallow water bodies.
- Distribution: found throughout India. Extinct in Bhutan.
- Threats: Habitat destruction because of conversion of natural habitats for agricultural and industrial use, sacrificed in superstitious practices or used as aphrodisiacs (any food or drug that arouses sexual instinct).
- IUCN: Vulnerable | CITES: Appendix I | Wildlife Protection Act: Schedule I

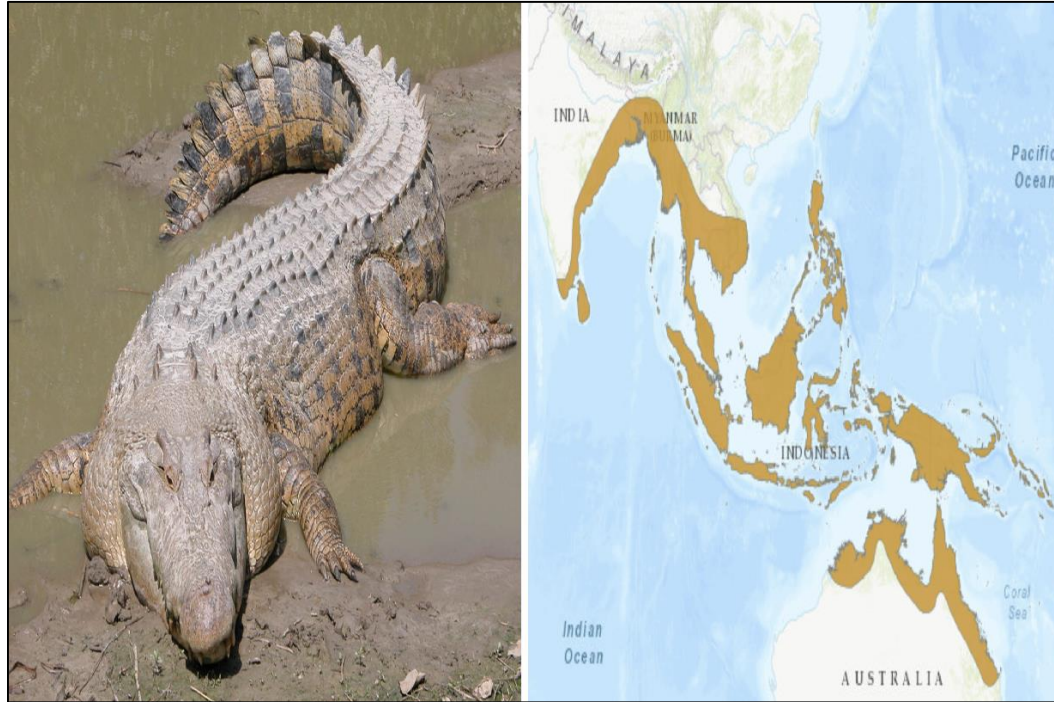


Saltwater crocodile

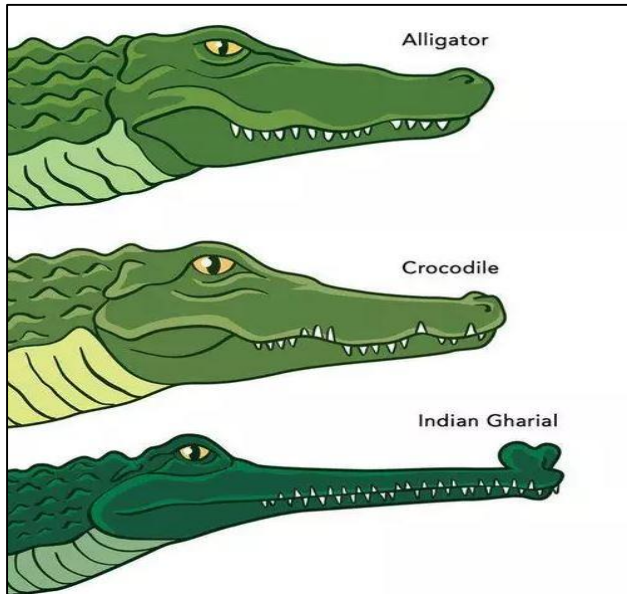


- Saltwater Crocodile is the largest living reptile and crocodilian known to science.
- Habitat: saltwater habitats and brackish wetlands
- Distribution: India's east coast (Odisha's Bhitarkanika Wildlife Sanctuary, Andaman and Nicobar Islands coasts and Sundarbans), across Southeast Asia and the Sundaic region to northern Australia.
- Threat: Hunted for its skin, illegal killing and habitat loss.
- IUCN: Least Concern | CITES: Appendix I | Wildlife Protection Act: Schedule I and Schedule IV

Saltwater crocodile



Crocodiles- comparison



Characteristics	Crocodiles	Alligators	Gharials
Family	<i>Alligatoridae</i>	<i>Crocodylidae</i>	<i>Gavialidae</i>
Jaw/Snout Shape	long and pointed V-shaped snouts	round and U-shaped snouts	long and pointedly thin snouts
Teeth Placement	Upper and lower jaws are of similar width	A wider upper jaw that overlaps the lower	With large front teeth for grasping, the lower jaw is slightly narrower than the upper
Water Body	Saltwater	Freshwater	Freshwater
Length of Body	Can grow up to 17 feet	Can grow up to 14 feet	Can grow up to 15 feet
Weight	Up to 2200 lbs (997.9 kg)	Up to 1000 lbs (453.5 kg)	Up to 2000 lbs (907.18 kg)

River dolphins



- **Ganges River Dolphin**
- **Characteristics:**
 - Can only live in freshwater and is blind.
 - Has a slit similar to a blowhole on top of its head, which acts as a nostril.
 - Popularly referred to as 'Susu' due to the sound it produces when breathing.
 - Symbols of the ecological health of our major river systems.
- **Threats:**
 - Direct killing, habitat fragmentation by dams and barrages and indiscriminate fishing.

A	<i>Common Name</i> Susu
Aa	<i>Scientific Name</i> <i>Platanista gangetica gangetica</i>
	<i>Population</i> Less than 1800 (1200 to 1800)
	<i>Length</i> Up to 2.70 m (Female), 2.12 m (Male)
	<i>Weight</i> 150-170 Kg.
	<i>Status</i> Endangered (IUCN)

River dolphins



- **Habitat**
- Ganges river dolphins prefer deep waters, in and around the confluence of rivers.
- Covers seven states namely, Assam, Uttar Pradesh, Madhya Pradesh, Rajasthan, Bihar, Jharkhand and West Bengal.
- The upper Ganga River (in Uttar Pradesh), Chambal River (Madhya Pradesh and Uttar Pradesh), Ghaghra and Gandak Rivers (Bihar and Uttar Pradesh), Ganga River,
- From Varanasi to Patna (Uttar Pradesh and Bihar), Son and Kosi rivers (Bihar)
- Brahmaputra from Sadia (foothills of Arunachal Pradesh) upto Dhubri (on the Bangladesh border) and Kulsi River, a tributary of the Brahmaputra River, form ideal habitats for the Ganges river dolphin.
- Vikramshila Gangetic Dolphin Sanctuary (VGDS) in Bihar is India's only sanctuary for the Gangetic dolphin.

River dolphins



River dolphins



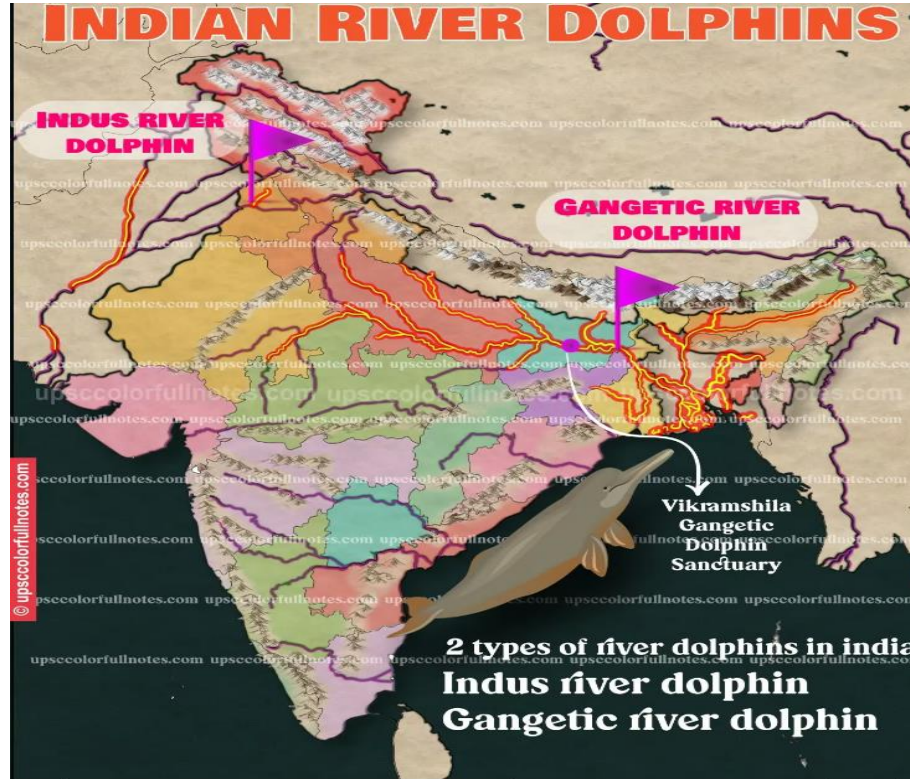
- **Indus River Dolphin**
- It is the State Aquatic Animal of Punjab.
- Also called Bhulan (Ganges river dolphin- Susu)
- They are **freshwater**, and **functionally blind** species of dolphins which rely on echolocation to navigate, communicate and hunt prey including prawns, catfish and carp.
- (Echolocation occurs when an animal emits a sound wave that bounces off an object, returning an echo that provides information about the object's distance and size.)
- Except for a tiny, isolated population of about 30 in India's Beas River (185 km stretch between Talwara and Harike), Indus river dolphins live exclusively in the Indus river in Pakistan.

River dolphins

- In 2017, a survey was done by WWF-Pakistan which showed an increase in their population,
- Similar survey is being conducted in India now with the help of WWF-India.
- The Punjab government along with WWF-India are conducting the first organised census on population of Indus Dolphin.
- IUCN Status- Endangered| Wildlife Protection Act- Schedule 1



River dolphins



River dolphins

INDUS RIVER DOLPHIN

- Indus river dolphins are believed to have **ORIGINATED IN THE ANCIENT TETHYS SEA.**

Today, they can only be found in the

- lower parts of the **INDUS RIVER IN PAKISTAN AND IN RIVER BEAS**

They have adapted to **LIFE IN THE MUDDY**

- **RIVER AND ARE FUNCTIONALLY BLIND.**

- They rely on **ECHOLLOCATION TO NAVIGATE**

- **STATE AQUATIC ANIMAL OF PUNJAB**

- IUCN status: **ENDANGERED.**

- They live for at least for **30 YEARS** and grow over **2 METRES IN LENGTH.**

GANGETIC RIVER DOLPHIN

- Gangetic River Dolphin is primarily found in the **GANGES AND BRAHMAPUTRA RIVERS.**

- They **PREFER DEEP WATERS** in and around the confluence of rivers.

- They also act as an **INDICATOR OF THE HEALTH OF THE FRESHWATER** ecosystem as they can only live in freshwater.

- Gangetic River dolphin is **INDIA'S NATIONAL AQUATIC ANIMAL.**

- IUCN Status: **ENDANGERED**

CITES: Appendix I

WPA1972: Schedule 1.

River dolphins

- **Irrawaddy Dolphin**
- They live in both salt- and freshwater in South and Southeast Asia.
- Three exclusively freshwater populations are found in Irrawaddy/Ayeyarwady River (Myanmar), Mekong River (Lao PDR, Cambodia); and Mahakam River (Indonesia).
- In addition, very small numbers survive in the partially freshwater Songkhla Lake (Thailand) and the brackish Chilika Lake (India).
- During the Dolphin Census in Chilika lake, direct sighting of 146 Irrawaddy dolphins has been reported.
- Irrawaddy dolphin is the only salt water dolphin found in India.
- IUCN Status- Endangered

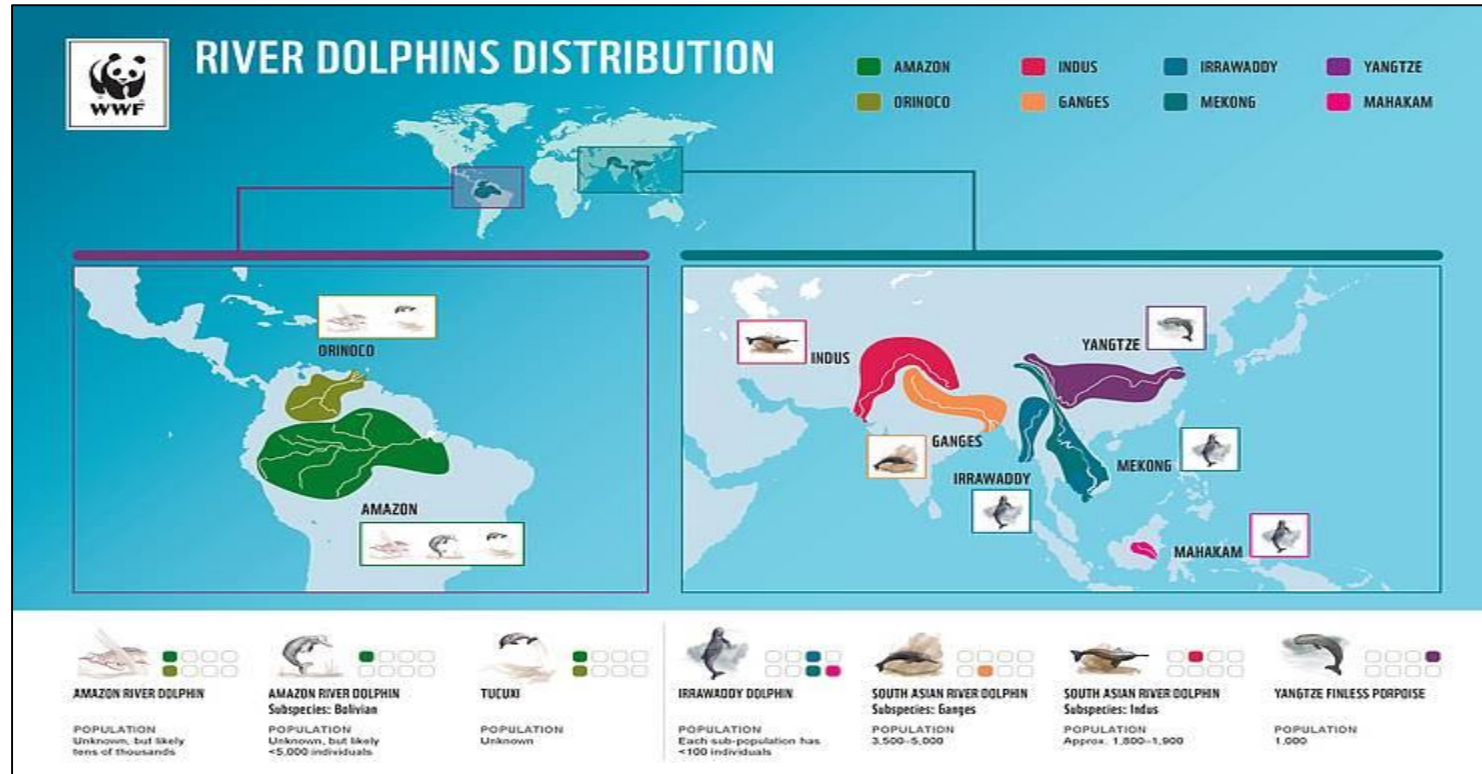


River dolphins



- River Dolphins in the world
 - Besides above 3 river dolphins in India, and 4 other important species across the world
1. Amazon River Dolphin (the pink river dolphin or Boto) are found only in freshwater and are Vulnerable.
 2. Tucuxi (Data Deficient Category), found in Amazon and its tributaries, can live in both salt- and freshwater.
 3. Yangtze river dolphin (Baiji) of China has been declared "functionally extinct" in 2006.
 4. Yangtze/Finless Porpoise (only, porpoise species that can live in freshwater) is Endangered and is found in the Yangtze River and its adjacent lake systems.

River dolphins



Red Panda

- **Characteristics:**
- Shy, solitary and arboreal (spending life on trees) animals.
- An indicator species for ecological change.
- Feed mainly on bamboo and has an extra thumb for feeding and climbing.
- Use their long, bushy tails for balance and to cover themselves in winter .



Red Panda

- **Habitat:**
- Almost 50% of red panda's habitat is in the Eastern Himalayas.
- In India, found in Sikkim (State Animal), Arunachal Pradesh, West Bengal and Meghalaya.
- India is home to both (sub) species: Himalayan red panda (*Ailurus fulgens*) and Chinese red panda (*Ailurus styani*)
- **Threats:**
- Habitat loss and degradation, human interference and poaching.



Red Panda



Dugong

- **Characteristics:**

- Also known as Sea Cows.
- Only herbivorous marine mammals and feed on seagrass.
- Only member of the Order Sirenia found in India.



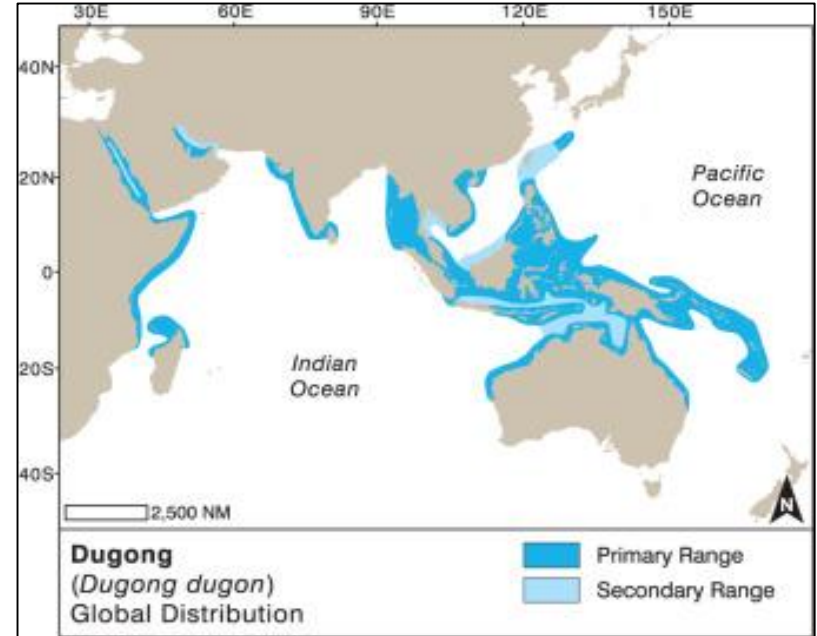
- Live in groups and come to the surface to breathe with a distinct dolphin-like tail.
- Have mammary glands.

- **Habitat and distribution:**

- Shallow coastal waters of the Indian and western Pacific Oceans.
- In India, found in Gulf of Mannar, Palk Bay, Gulf of Kutch and Andaman and Nicobar Islands.
- State animal- Andaman and Nicobar islands



Dugong



- **Dugong Conservation Efforts in India**

- In 2022, the government of Tamil Nadu created India's first dugong conservation reserve in the Gulf of Mannar and the surrounding Palk Bay on the southeast coast of the country.

- **Threats:**

- Ocean floor trawling is one of the most important factors behind decreasing dugong populations in many parts of the world.
- Trawling is a method of fishing that involves pulling a fishing net through the water behind one or more boats.
- Destruction and modification of habitat, pollution, rampant illegal fishing activities, vessel strikes, unsustainable hunting or poaching and unplanned tourism are the main threats to dugongs.
- Dugong meat is consumed under the wrong impression that it cools down human body temperature.

Vulture Conservation



- India is the most favourable region for vultures.
- Hindus do not eat cows, and when a cow dies, it is left to be fed on by vultures.
- India also has high species diversity, and hence vultures get a lot of food.
- Nine species of vulture can be found in India. Most are in danger of extinction due to a veterinary drug called diclofenac (vultures do not have a particular enzyme required to break down diclofenac). Similar drug : **Aclofenac**.
- Diclofenac is a common nonsteroidal anti-inflammatory drug (NSAID – works like steroids, but without as many side effects) administered to livestock and is used to treat the symptoms of inflammation, fevers and/or pain associated with disease or wounds.
- The population of the White-Rumped Vulture fell 99.7% between 1993 and 2002.
- The populations of the Indian Vulture and the Slender-Billed Vulture fell 97.4%.

Vulture Conservation



- Consequence of population decline:
- Vultures played an important role in public sanitation.
- The carcasses formerly eaten by vultures' rot in village fields leading to contaminated drinking water.
- Their disappearance has resulted in an explosion of rats and wild dogs and the spread of diseases
- Link food chain, decomposers concepts

Vulture Conservation



Vulture	IUCN Status
White- rumped	Critically Endangered
Slender-billed	Critically Endangered
Indian	Critically Endangered
Red-headed	Critically Endangered
Egyptian	Endangered
Eurasian Griffon	Least Concerned
Himalayan Griffon	Near Threatened
Cinereous	Near Threatened
Bearded/ Lammergeier	Near Threatened

Vulture Conservation



- **Vulture Action Plan 2020-25:**
- Drug control: To ensure minimum use of Diclofenac
- Upscaling conservation: To establish Additional Conservation Breeding Centres along with Vulture Conservation Centres
- Eight Vulture Conservation Breeding Centres (VCBC) are established in the country. The first VCBC was set up in 2004 at Pinjore in Haryana
- Implement Vulture Safety Zone Programme at eight different places in the country where there are existing populations of vultures.
- Aim of developing VSZs is to establish targeted awareness activities surrounding 150 km radius of vultures' colonies so that no diclofenac or the veterinary toxic drugs are found in cattle carcasses.

Species in News

- **Musk Deer**
- Himalayan musk deer (or White-bellied musk deer) is a musk deer species occurring in the Himalayas. State animal- Uttarakhand
- Musk deer family differs from true deer (cervids) by lacking antlers.
- Male musk deer secretes a scented substance called musk which is used in perfume industry.
- Distribution: Himalayas of Bhutan, India, Nepal, and China.
- Habitat: High alpine environments.
- Threats: Hunting (for musk) and habitat loss.
- IUCN: Endangered | CITES: Appendix I | Wildlife Protection Act: Schedule I



Species in News

- **Nilgiri Tahr**
- Nilgiri Tahr, locally known as Varaiaadu, is the state animal of Tamil Nadu. It is the only mountain ungulate (animal with hooves) in southern India among the 12 species found in India. State animal- Tamil Nadu
- Ungulate: herbivorous hoofed mammal. E.g., cow.
- Distribution: It is endemic to the Nilgiri Hills. It is limited to the Western Ghats in Kerala and Tamil Nadu
- The Eravikulam NP in Anamalai hills (Kerala) has the largest population.
- Habitat: Tropical montane grasslands, sholas forests and rocky areas at high elevations .
- Threats: acacia and eucalyptus plantations.
- IUCN: Endangered | Wildlife Protection Act: Schedule I



Species in News

- **Nilgiri Tahr**
- only tahr is adapted to a cold and wet tropical environment.
- Diurnal, but most active in early morning and late afternoon.
- Endemic to Western Ghats, current distribution between niligiris in north and kanyakumaris in south.
- Mukurthi NP : created to protect the keystone species.

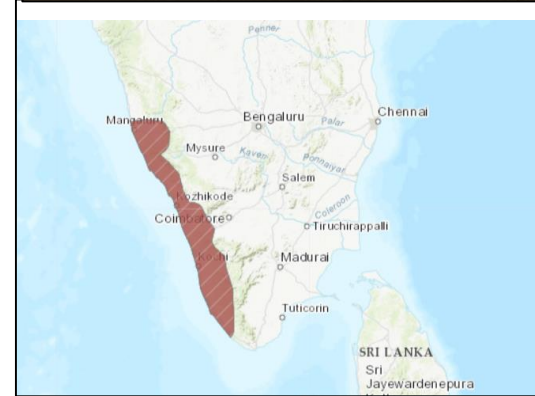


- **Pygmy Hog**
- Pygmy Hog is the world's smallest wild pig. It is one of the indicators species of grassland habitats.
- Pygmy hog-sucking Louse is a critically endangered parasite that feeds only on Pygmy Hogs.
- Distribution: This rare species of pig is found only in Manas WLS (Assam). Earlier, it was found across north India.
- Habitat: Dense tall riverine grasslands.
- Threats: Habitat loss and hunting.
- IUCN: Endangered | Wildlife Protection Act: Schedule I



Species in News

- **Malabar Civet**
- It is a nocturnal viverrid endemic to the Western Ghats of India.
- **Distribution:** Malabar civet occurred throughout the Malabar coast from the latitude of Honnavar to Kanyakumari.
- **Threats:** Habitat loss and fragmentation..
- **IUCN: CR**

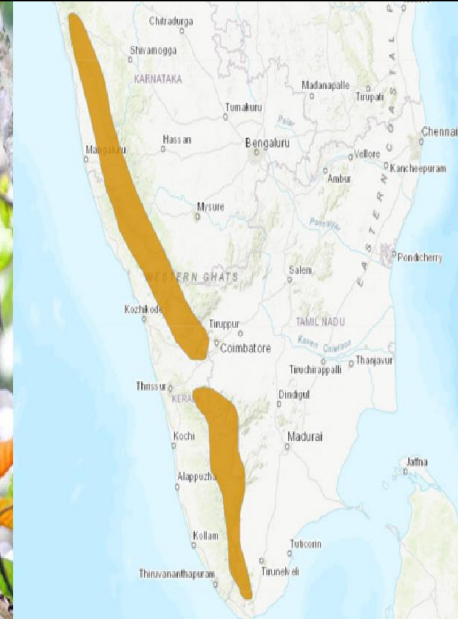


- **Kashmir stag or Hangul**
- Only Asiatic sub-species of the Elk (the largest subspecies of Red deer, found in high mountains of Central Asia).
- It is the state animal of J&K.
- Distribution: It is endemic to Kashmir and is now restricted to the Dachigam NP.
- Habitat: Mountainous areas, where it spends summers in alpine meadows and winters in valleys.
- Threats: over-grazing by domestic livestock and poaching.
- IUCN: Endangered | CITES: Appendix I | Wildlife Protection Act: Schedule I | CMS: Appendix I



Species in News

- **Lion-tailed macaque or wanderoo**
- Old World monkey endemic to the Western Ghats, India.
- Unlike other macaques, this arboreal species avoids humans.
- Distribution: Karnataka, Kerala, and TN.
- Habitat: Prefers tropical evergreen rainforests but also found in monsoon forests.
- Threats: Habitat loss, hunting and wood harvesting.
- IUCN: Endangered | CITES: Appendix I | Wildlife Protection Act: Schedule I

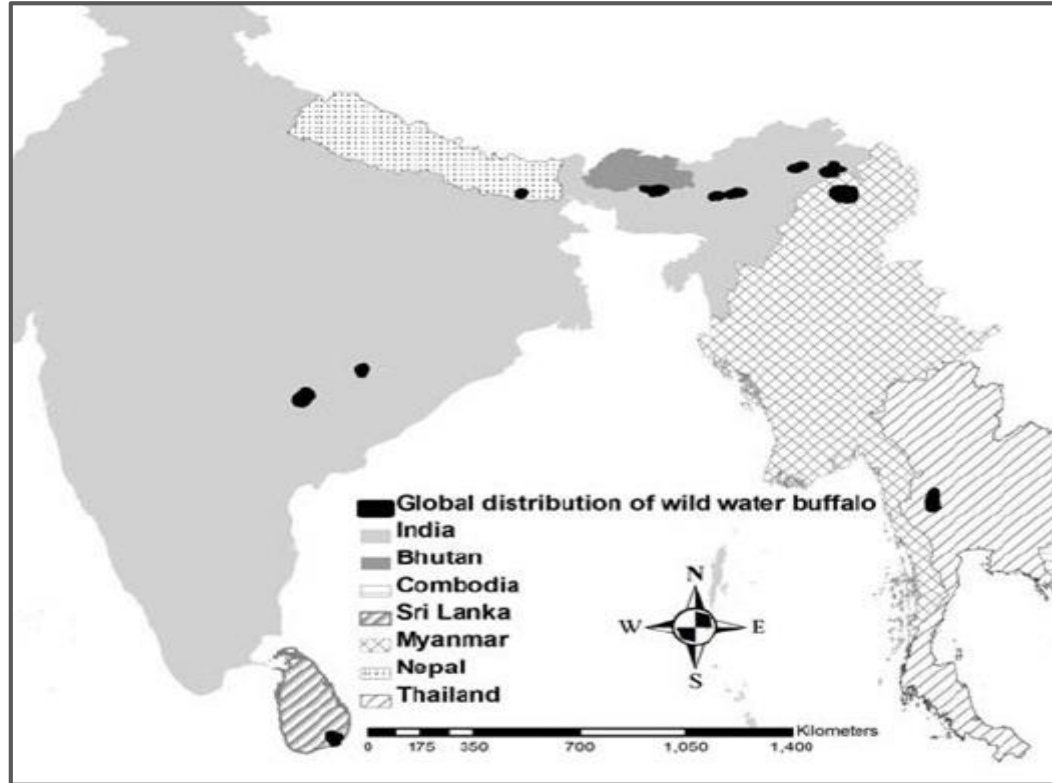


Species in News

- **Wild water buffalo/ Asiatic buffalo/ Wild buffalo**
- Large bovine native to the Indian subcontinent & SE Asia.
- It is the state animal of Chhattisgarh.
- Distribution: South & SE Asia. In India, it is found in the Central & NE India, predominantly in Assam.
- Kaziranga, Manas & Dibru-Saikhowa NPs are famous for it.
- Habitat: Low-lying alluvial grasslands including beels (ox-bow), river banks, and chaporis (small sandy islands within braided river systems).
- IUCN: Endangered | CITES: Appendix III | Wildlife Protection Act: Schedule I



Species in News



Species in News

- **Sangai**
- Sangai (Brow-antlered deer or Dancing deer) is a subspecies of Eld's deer (found in South and SE Asia).
- It is endemic to Manipur. It is the state animal of Manipur.
- Distribution: Found only at Keibul Lamjao NP over the floating biomass in the Logtak Lake locally called phumdi.
- Threats: Habitat loss and hunting.
- IUCN: Endangered | CITES: Appendix I | Wildlife Protection Act: Schedule I

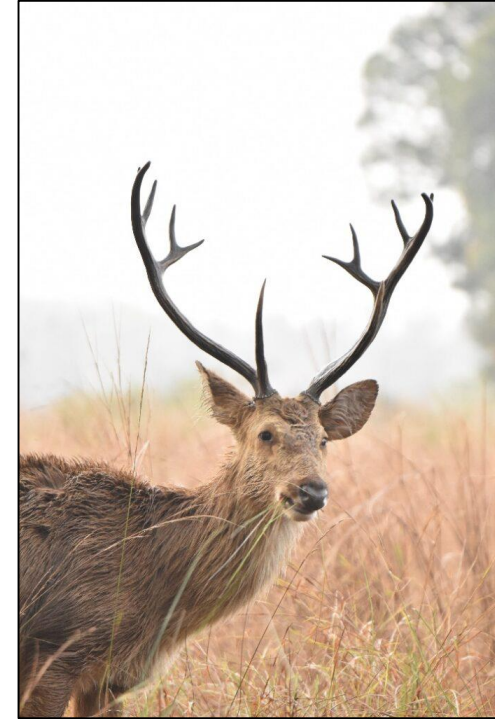


Species in News

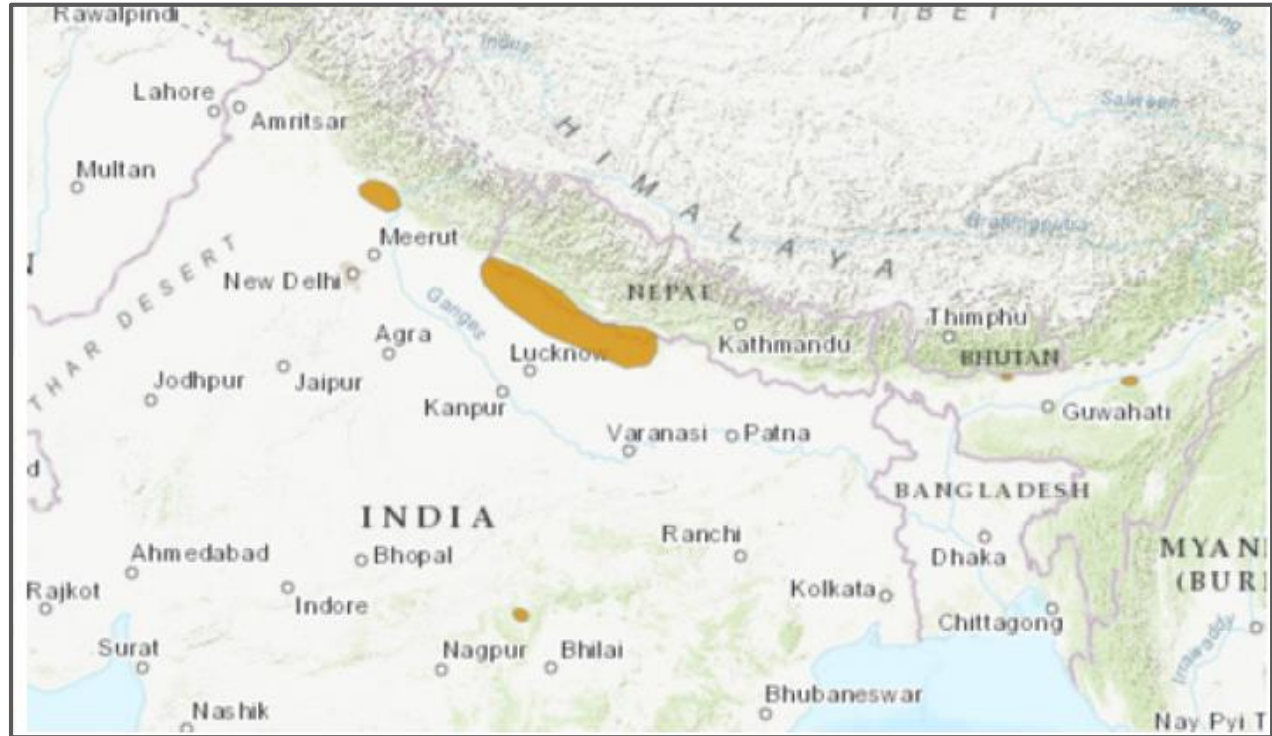


Species in News

- **Barasingha or Swamp deer**
- A deer species native to the Indian subcontinent.
- It is the state animal of Madhya Pradesh and Uttar Pradesh.
- Distribution: Isolated localities in north and central India, and southwestern Nepal.
- Kanha National Park(MP) is famous for it.
- Habitat: Open forests, tall grasslands and mangroves.
- IUCN: Vulnerable| CITES: Appendix I | Wildlife Protection Act: Schedule I



Species in News



- **Clouded Leopard**
- Clouded leopard is a wild cat inhabiting dense forests.
- It is the state animal of Meghalaya.
- Dampa Tiger Reserve (Mizoram) has the highest population density of clouded leopards.
- Distribution: Bangladesh, Bhutan, China, India, Nepal and SE Asia. In India, it occurs upto Valmiki TR in Bihar.
- IUCN: Vulnerable | CITES: Appendix I | Wildlife Protection Act: Schedule I



Species in News

- **Slender Loris**
- Context- India's first sanctuary for slender lorises was notified recently, named Kadavur Slender Loris Sanctuary, in Tamil Nadu.
- Arboreal in nature i.e., spend most of their life on trees.
- Primates that cannot jump.
- Threats: Threat from poachers due to the misplaced belief that these animals have magical and medicinal powers, destruction of their habitat.
- Habitat: Commonly found in the tropical scrub and deciduous forests and the dense hedgerow plantations bordering farmlands of Southern India and Sri Lanka.
- Grey Slender Loris: found only in South India and small parts of Sri Lanka. IUCN Status: Near Threatened
- Red Slender Loris: found only in Sri Lanka. IUCN Status- Endangered



Species in News

- **Fishing Cat**
- Fishing cat is a medium-sized wild cat of South and SE Asia.
- This nocturnal cat is an adept swimmer and prey on fish.
- It is the state animal of West Bengal.
- Distribution: In India, it is found in plains of Ganga, Yamuna, Brahmaputra, Sundarbans delta, coastal wetlands along the Bay of Bengal, and Chilika lagoon.
- Habitat: Wetlands, mangroves, rivers and streams.
- Threats: Habitat loss (esp. due to shrimp farming), unsustainable fishing practices and poaching.
- IUCN: Vulnerable | CITES: Appendix II | Wildlife Protection Act: Schedule I



Species in News



Species in News

- **Rusty Spotted Cat**
- One of the smallest cats.
- Found in India except West Bengal and NorthEast.
- Absent in Evergreen Forests.
- IUCN: NT | CITES: Appendix I | Wildlife Protection Act: Schedule I



Species in News

- **Gaur or Indian bison**
- Largest among the wild cattle and the bovids.
- Domesticated form of the gaur is called Gayal or Mithun.
- It is an attraction of National Parks of Wayanad, Nagarhole, Mudumalai & Bandipur.
- Trishna WLS in Tripura is the India's only natural breeding centre of bisons. State animal- Bihar, Goa
- Distribution: It is native to South Asia, SE Asia and East Asia. In India, it is found in parts of Himalayan foothills, NE India, Eastern Ghats and Western Ghats.
- Habitat: Evergreen, semi-evergreen, moist deciduous and dry deciduous forests.
- IUCN: Vulnerable| CITES: Appendix I | Wildlife Protection Act: Schedule I



Species in News

- **Indian Wild Ass**
- The Indian wild ass or the Indian onager is a subspecies that belongs to the onager family and is native to Southern Asia
- **Threats: Illegal salt panning activities, Invasive shrubs in their habitat result in the loss of food, Uncontrolled grazing**
- The Indian Wild Ass Sanctuary is located in the little Rann of Kutch in Gujarat
- IUCN Status- Near Threatened



Species in News



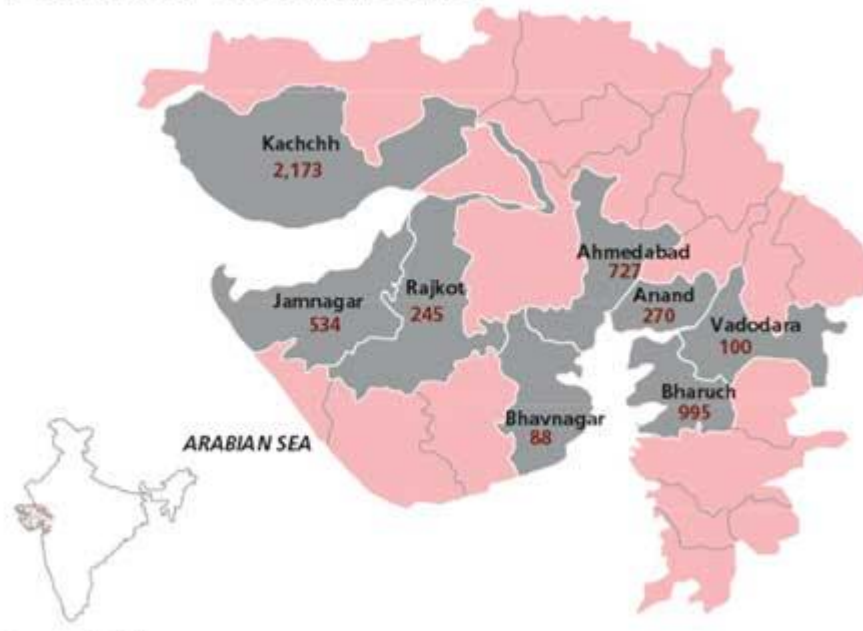
Species in News

- **Kharai Camel**
- Their name is derived from the local word khara, meaning saline.
- Location: Kutch, a coastal region of Gujarat, which is also a large desert land, has two camel breeds. One is the popular Kutchi breed and the other, the Kharai breed, native to the region.
- The Kharai breed has the special ability to survive on both dry land and in the sea, making it an ecotonal breed.
- Recognised as a separate breed a few years, the Kharai camel is probably the only domesticated breed of camel that lives in dual ecosystems.
- These camels have a special ability to swim in seawater and feed on saline plants and mangroves, which is how they get their name, Kharai ('salty' in Gujarati).
- IUCN Status- Endangered



Distribution of Kharai Camel

Population of Kharai camel



Species in News

- **Pangolin (Scaly Anteater)**
- A nocturnal scaly skinned insectivorous mammal.
- This burrow dwelling animal is the most trafficked mammal in the world.
- Rolls up and protects its vulnerable parts when attacked by a predator
- Threats: Trafficking and hunting (for its meat and scales), habitat loss (esp., mining & quarrying, agriculture) etc.
- Out of the eight species of pangolin, Indian Pangolin & Chinese Pangolin are found in India.
- Chinese Pangolin - Critically Endangered
- India Pangolin- Endangered



Species in News



Distribution of Indian Pangolin



Distribution of Chinese Pangolin

Species in News

- **Kashmir gray langur** (*Semnopithecus ajax*)
- Habitat- It has been reported from Jammu and Kashmir and Himachal Pradesh in northwestern India but evidence indicates it only occurs in the Chamba Valley in Himachal Pradesh
- Threat- restricted range, fragmented population , overgrazing, building roads through forests, logging, deforestation, agriculture, and fire
- IUCN Status- Endangered | CITES- Appendix I | Wildlife Protection Act- Schedule I



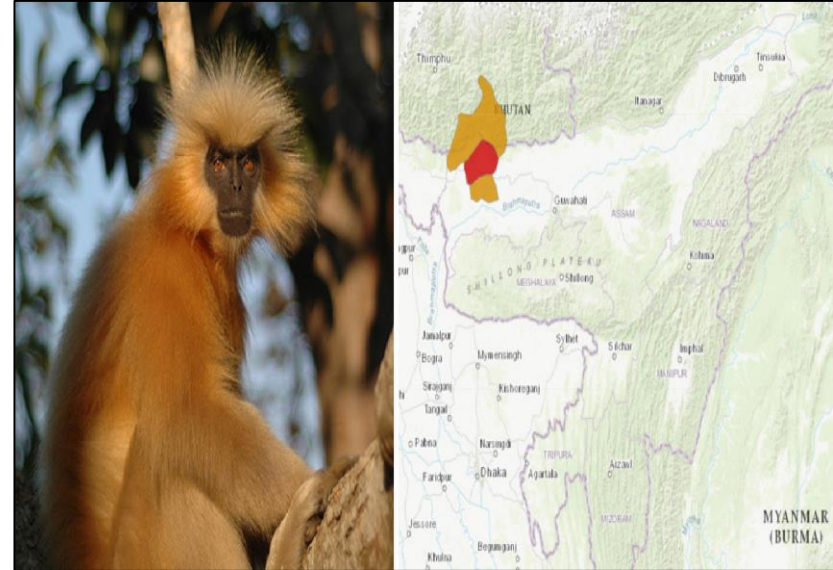
Species in News

- **Phayre's Leaf Monkey**
- Habitat- northeastern India (Assam, Mizoram, and Tripura)
- State animal- Tripura
- Threat- hunting and loss of habitat due to jhum (slash and burn) cultivation, timber logging
- IUCN Status- Endangered | Wildlife Protection Act- Schedule I



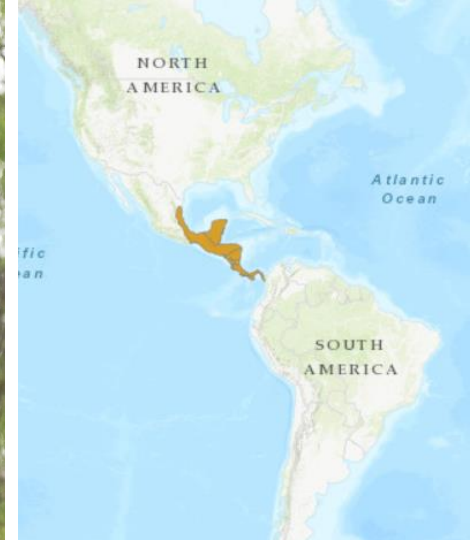
Species in News

- **Gee's golden langur or Golden langur**
- An Old World (native to Africa and Asia) monkey with golden fur.
- **Distribution:** It is endemic to Western Assam and Southern Bhutan. Manas NP (India), Royal Manas NP (Bhutan) and Black Mountains (Bhutan) are famous for the species.
- **Habitat:** Moist evergreen, riverine, moist deciduous forests, temperate and subalpine forests.
- **Threats:** Inbreeding depression.
- **IUCN Status-** Endangered | **Wildlife Protection Act-** Schedule I



Species in News

- **Spider Monkey**
- Spider monkeys are New World monkeys belonging to the genus *Ateles*, part of the subfamily Atelinae, family Atelidae. Spider monkeys are frugivorous (eat fleshy fruits 71 to 83 percent of the time).
- **Distribution:** They are found in tropical forests of Central and South America, from southern Mexico to Brazil.
- **Habitat:** Live in the upper layers of the rainforest, and forage in the high canopy,
- **IUCN Status:** Endangered



Species in News

- **Kondana Rat**
- Kondana rat is a nocturnal burrowing rodent which is found only in Pune, Maharashtra, India.
- **Distribution:** It is known to occur at only four localities of Pune, namely, the Singharh plateau (Singharh was earlier known as Kondana), the Torna plateau, the Rajgad plateau, and Rareshwar.
- **Habitat:** Tropical and subtropical dry deciduous forests.
- **IUCN:** EN



- **Andaman White Toothed Shrew**
- Andaman white toothed shrew is a small mouse-like insectivorous nocturnal mammal. It was recently discovered in Narcondam Island, a volcanic island.
- **Distribution:** It is endemic to the Andaman Islands.
- **Habitat:** Tropical moist deciduous and evergreen forests.
- **Threats:** Natural disasters (tsunami, earthquake, flooding, storms, etc.)
- **IUCN Status:** CR



Species in News

- **Elvira rat**
- Large rock rat (or Elvira rat) is a medium-sized, nocturnal and fossorial (burrowing) rodent.
- **Distribution:** It is endemic to type locality of Kurumbapatti, Salem District (**Eastern Ghats**, TN).
- **Habitat:** Rocky areas of tropical dry deciduous scrub forest.
- **Threats:** Habitat loss (esp. due to mining and quarrying, logging, agriculture)
- **IUCN: CR**



Species in News

- **Namdapha Flying Squirrel**
- Nocturnal like all other flying squirrels.
- It is restricted to Namdapha NP (the largest protected area in the Eastern Himalaya biodiversity hotspot)
- Habitat: Along streams of dry deciduous montane forests.
- Threats: Habitat loss and climate change.
- IUCN Status- Critically Endangered
- Wildlife Protection Act- Schedule I



Species in News

- **Indian Giant Squirrel**
- Large multi-coloured tree squirrel species endemic to forests and woodlands in India.
- It is a diurnal, arboreal, and mainly herbivorous squirrel
- Distribution- This species is endemic to India, with main sections of its distribution in the Western Ghats, Eastern Ghats and Satpura Range as far north as Madhya Pradesh
- Habitat: tropical deciduous, semi-deciduous (where often utilizing denser riparian growth), and moist evergreen forests and woodlands.
- State animal- Maharashtra
- IUCN Status- Least Concern| Wildlife Protection Act- Schedule I



Species in News

- **Himalayan Brown Bear**
- It is one of the largest carnivores in the highlands of Himalayas.
- It is omnivorous and it hibernates during the winter.
- Habitat: Forests at high altitudes and alpine meadows.
- Distribution: Nepal, Tibet, north India, and north Pakistan.
- Threats: loss of suitable habitat and persecution by humans
- While the brown bear as a species is classified as Least Concern (IUCN), this Himalayan sub-species is highly endangered, and population is declining
- IUCN- Endangered | Wildlife Protection Act- Schedule I



Species in News

- **Blackbuck**
- Blackbuck (Indian antelope) is an antelope native to India and Nepal.
- It is the state animal of Andhra Pradesh, Haryana, Punjab
- Distribution: Formerly, found in almost whole of the Indian subcontinent south of the Himalayas. At present, it is extinct in Bangladesh and Pakistan.
- Habitat: Grasslands, dry thorn scrubs, scrublands, lightly-wooded country and agricultural margins.
- Threats: Hunting and habitat loss
- IUCN- Least Concern | Wildlife Protection Act- Schedule I



Species in News

- **Himalayan Serow**
- It is mostly blackish, with flanks, hindquarters, and upper legs that are a rusty red; its lower legs are whitish
- Habitat: hilly forests above an elevation of 300 m (980 ft), but descends to 100 m (330 ft) in winter.[8] It prefers elevations of 2,500–3,500 m (8,200–11,500 ft) in the Himalayas.
- State animal- Mizoram
- IUCN- Near Threatened| Wildlife Protection Act- Schedule I



Species in News

- **Chinkara or Indian gazelle**
- It is native to Iran, Afghanistan, Pakistan & India.
- It is the smallest Asiatic antelope.
- There are six subspecies of Indian gazelle. Of these Deccan chinkara and Gujarat chinkara are found in India.
- Distribution: The Deccan chinkara ranges from the Ganges Valley (west of WB) to Deccan Plateau. The Gujarat chinkara are found in Thar Desert, Rann of Kutch, Kathiawar, and Saurashtra region.
- State animal- Rajasthan
- Habitat: Arid, deserts, dry scrub and light forests.
- It is a shy species and avoids human habitation.
- IUCN- Least Concern | Wildlife Protection Act- Schedule I



Species in News

- **Chital**
- Also called Spotted deer, is a deer species native to the Indian subcontinent.
- Antlers are present only on males.
- Distribution: In India, it is not found in high Himalayan regions and arid areas of Northwest India.
- Habitat: Thrives in a variety of habitats, but avoids extremes such as dense evergreen forests and desert.
- State animal- Telangana
- IUCN- Least Concern | Wildlife Protection Act- Schedule II



Species in News

- **Sambar**
- Sambar is a large deer species native to the Indian subcontinent and SE Asia.
- This nocturnal species is the state animal of Odisha.
- Distribution: India (except hot deserts of Northwest and high altitude Himalayas), Nepal, Bhutan & SE Asia.
- Habitat: No large Indian ungulate has adapted itself to a wider variety of forest types than Sambar. However, it is always found close to water source.
- IUCN- Vulnerable | Wildlife Protection Act- Schedule I
- State animal- Odisha



Species in News

- **Nilgai**
- Also called blue bull, it is the largest antelope of Asia and found in India, Nepal and Pakistan.
- State animal- Delhi
- Distribution: Indravati NP (Chhattisgarh), Pench TR (MP), Ranthambhore NP and Keoladeo NP are famous for it.
- Habitat: Arid areas, grassy plains, dry deciduous open forests & farmlands. Avoids dense forests and deserts.
- IUCN- Least Concern | Wildlife Protection Act- Schedule II

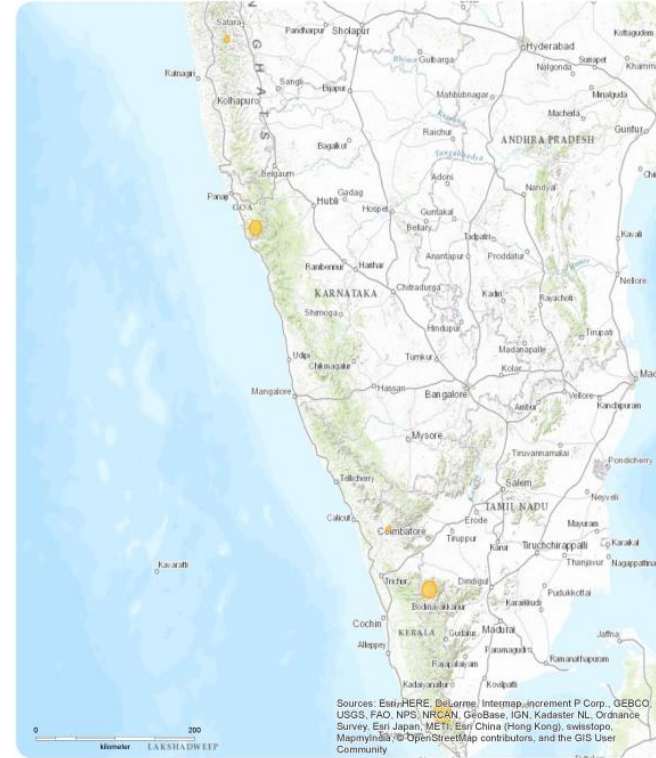


Species in News

- **Malabar Tree Toad**
- The Malabar tree toad or warty Asian tree toad, is a species of toad found in forests along the Western Ghats of great Karnataka or Deccan.
- **It is a small species and is found in wet tree hollows or leaf bases containing water.**
- **Habitat:** They are found in the forest of the Western Ghats at elevations of 250 m to over 1000 m, often beside streams. **Protected Areas:** Mhadei Wildlife Sanctuary, Mollem NP
- **IUCN Status: CE**



Distribution of Malabar Toad



Sources: Esri, HERE, DeLorme, Intermap, increment P Corp., GEBCO, USGS, FAO, NPS, NRCAN, GeBCo, IGN, Kadaster NL, Ordnance Survey, Esri Japan, METI, Esri China (Hong Kong), swisstopo, Mapbox India, © OpenStreetMap contributors, and the GIS User Community

Species in News

- **Star Tortoise**
- **Habitat:** Parts of India below WB and Gujarat.
- **Threats:** traded for meat and pet trade.
- IUCN: Vulnerable | CITES: Appendix I | Wildlife Protection Act: Schedule I
- **Giant Leatherback turtle**
- Giant Leatherback turtles are named for their shell.
- Their shells are leather-like rather than hard, like other turtles.
- They are the largest of the seven species of sea turtles on the planet and also the most long-ranging.
- IUCN Status: Vulnerable
- Wildlife Protection Act, 1972: Schedule I



Species in News



Distribution of Leatherback turtle

Species in News

- **Olive Ridley turtle**
- Smallest and most abundant of all sea turtles.
- Best known for their unique mass nesting called Arribada, where thousands of females lay eggs on the same beach.
- A single turtle can lay over 110 - 140 eggs around.
- Diet: Carnivores, feed on jellyfish, shrimp, snails, crabs etc.
- Threats: Unintended capture in fishing gear, Direct Harvest of Turtles and Eggs, loss and degradation of habitat, marine debris, climate change etc.
- Habitat: Warm waters of the Pacific, Atlantic and Indian oceans.
- Rushikulya river mouth is considered the second-biggest rookery in India after Gahirmatha.
- IUCN- Vulnerable | Wildlife Protection Act- Schedule I



NESTING SITES OF OLIVE RIDLEY TURTLES



Species in News

- **Red-crowned roofed turtle**
- Freshwater turtle species, found in deep flowing rivers with terrestrial nesting sites.
- In comparison to their female counterparts, the males are shorter.
- Threats: Loss or degradation of habitat, drowning by illegal fishing nets, Poaching and illegal trade etc.
- Habitat: Native to India, Nepal and Bangladesh.
- Currently in India, the National Chambal River Gharial Sanctuary is the only area with substantial population of the species.
- IUCN- Critically Endangered | CITES- Appendix I | Wildlife Protection Act- Schedule I



Species in News

- **Assam Roofed Turtle**
- Habitat: north-eastern & south-eastern parts of Bangladesh, in India (Assam) and possibly in Bhutan
- Threats: It is exploited for its meat and eggs for local consumption or be collected for the pet trade especially in Asia. Habitat destruction by logging and incidental capture in fishing gear are also thought to present threats.
- IUCN Status- Endangered | Wildlife Protection Act- Schedule 1
- **Keeled Box Turtle**
- Habitat- mostly north east India
- Threat- Logging of rainforests for timber or agriculture, capture for the international pet trade, capture for food and traditional medicine
- IUCN Status- Endangered | Wildlife Protection Act- Schedule I



Species in News

- **Hawksbills turtle**
- Habitat- It is reported from Lakshadweep, southwest coast, Tamil Nadu and Andaman & Nicobar Islands.
- It feeds mainly on sponges, crabs and molluscs.
- Threat -hawksbills are threatened by the loss of nesting and feeding habitats, excessive egg collection, fishery-related mortality, pollution, and coastal development. However, they are most threatened by wildlife trade
- IUCN Status- Critically Endangered | CITES- Appendix I | Wildlife Protection Act- Schedule I



Species in News

- **Southeast Asian box turtle**
- They are omnivorous turtles with blackish-brown to olive-brown colored shells.
- Habitat: It is found in the Nicobar Islands, eastern India (Assam) and South East Asian Countries.
- **IUCN Status:** Endangered
- CITES: Appendix II

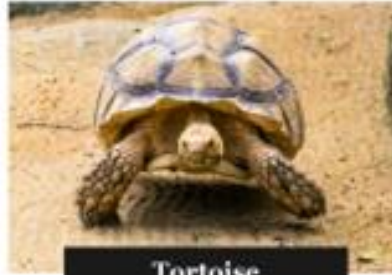


Turtle or Tortoise?



Turtle

- Flat, lightweight shell
- Webbed feet or flippers
- Omnivores
- Live primarily in water
- Shorter lifespan (20 – 40 years)



Tortoise

- Thick, domed shell
- Stumpy, clubbed feet
- Herbivores
- Live entirely on land
- Longer lifespan (60 – 80+ years)

- **Monitor Lizard**
- Large-sized lizards and have long necks, well-developed limbs and powerful tails and claws.
- Native to Africa, Asia, Oceania, and the Americas
- Monitor Lizards are carnivorous and consume insects, reptiles, birds, fishes etc
- Threats- traditional medicines, food and leather industry
- Bengal monitor (*Varanus bengalensis*), also called the common Indian monitor, is a monitor lizard distributed widely in the Indian Subcontinent, as well as parts of Southeast Asia and West Asia
- IUCN of Bengal monitor- Near Threatened



- **Blue Duke Butterfly**
- Context
- A native butterfly species to Sikkim was announced as the State Butterfly of Sikkim.
- Characteristics:
- Found at an altitude below 1,500 metres
- Discovered in Sikkim in 1858.
- Habitat: Native to Sikkim and the eastern Himalayas.
- Maharashtra was the first state to name a State Butterfly- Blue Mormon



Species in News

- **Great Indian Hornbill**
- Is the largest hornbill in India.
- Southern ground hornbill of Africa is the largest species of hornbill in the world.
- It is the state bird of Arunachal Pradesh and Kerala.
- Distribution: It is native to Indian Subcontinent, SE Asia and China. In India, it is found in North-east India and Western Ghats.
- Habitat: Tropical evergreen forests.
- IUCN: Vulnerable | CITES: Appendix I | Wildlife Protection Act: Schedule I
- Hornbill Festival is celebrated every year from 1st to 10th December in Kimasa (Heritage Village of Nagaland) to celebrate nature and traditions of Naga tribes.

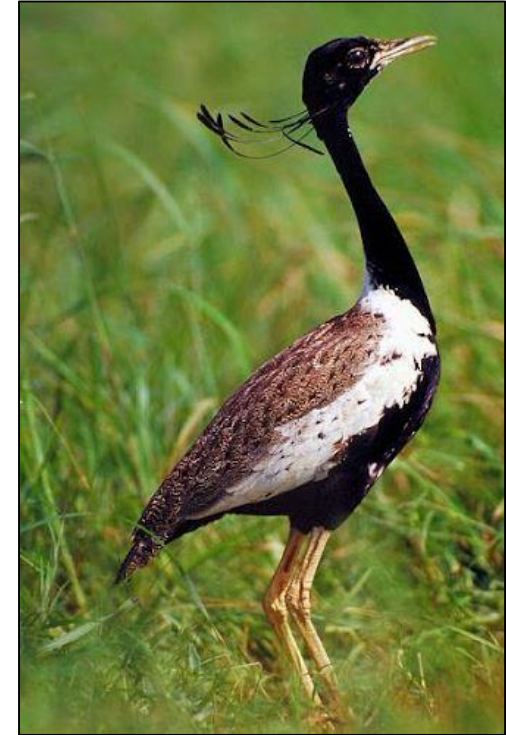


Species in News

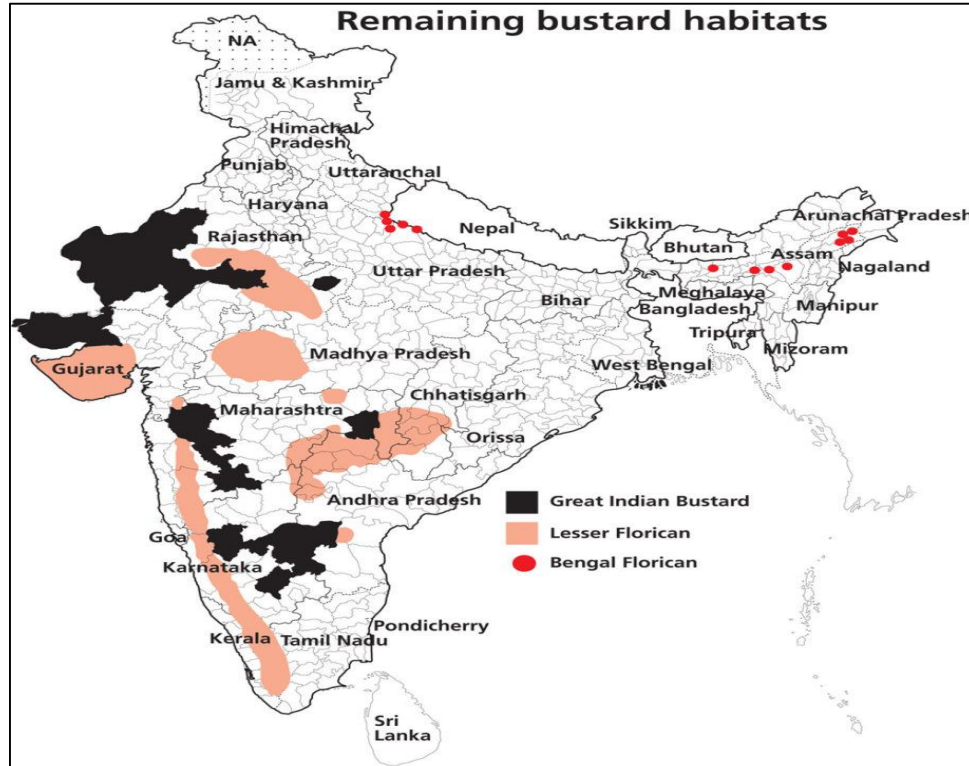


Species in News

- **Lesser Florican**
- Also known as Kharmor.
- A small and gracile bustard of the bustard family.
- Smallest of India's 3 resident bustards.
- Male floricans are generally shorter and lighter than their female counterparts.
- Threats: Hunting, Grassland conversion to agriculture, overgrazing, collision with energy infrastructure.
- Endemic to India and mainly found in grasslands and scrubby fields in parts of Rajasthan, Gujarat, Madhya Pradesh and Maharashtra.
- IUCN: Critically Endangered| CITES: Appendix II | Wildlife Protection Act: Schedule I



Species in News



Species in News

- **Black Necked Crane**
- Black necked crane is the state bird of UT of Ladakh (it was state bird of erstwhile J&K).
- Distribution: It is native to Ladakh (India), Sichuan (China) and Tibet. It migrates to Arunachal Pradesh (India) and Bhutan during winters.
- Habitat: Inland grassy wetlands at high elevation.
- IUCN: Near Threatened | CITES: Appendix I | CMS: Appendix I | Wildlife Protection Act: Schedule I



Species in News

- **Painted stork**
- It is a large shore bird endemic to Indian subcontinent and South East Asia.
- Distribution: River plains of Indian subcontinent and mainland SE Asia.
- Habitat: Freshwater marshes, lakes and reservoirs, flooded fields, river banks, etc.
- Threats: hunting, pollution, invasive species
- IUCN: Near Threatened



Species in News

- **Himalayan Quail**
- **Habitat:** Found only in the mountains of Uttarakhand in north-west Himalayas. Tall grass and scrub on steep hillsides.
- **Threats:** Hunting habitat modification – due to land use changes open cast mining for limestone and related disturbance
- Presumed to be extinct since no reliable records of sightings of this species exist after 1876. Possible sighting of this species was reported in Nainital in 2003
- **IUCN Status-** Critically Endangered | Wildlife Protection Act- Schedule I



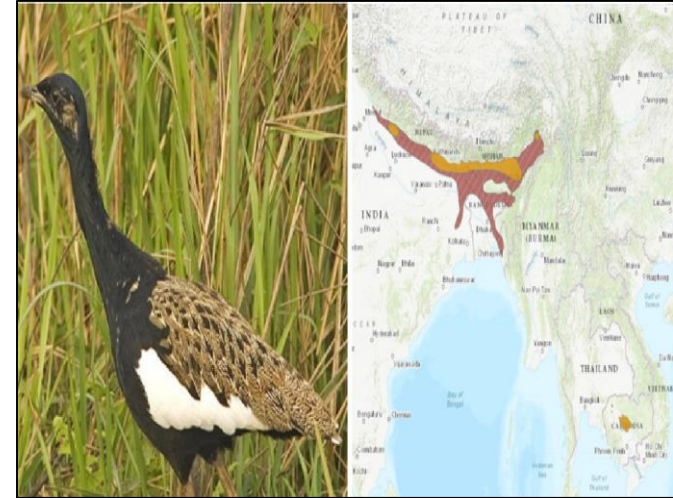
Species in News

- **Pink Headed Duck**
- It is a rare species of waterfowl that is last sighted in 1949 (it is feared to be extinct)
- It is unique in the pink colouration of the head combined with a dark body.
- The eggs have also been held as particularly peculiar in being nearly spherical.
- Distribution: It is recorded in India & Myanmar.
- Habitats: Wetlands surrounded by bushes and high grass.
- IUCN: Critically Endangered | CITES: Appendix I | Wildlife Protection Act: Schedule I



Species in News

- **Bengal Florican**
- It is a rare bustard species that is well known for its mating dance.
- Habitat: grasslands interspersed with scrublands.
- Distribution: native to the Indian subcontinent, Cambodia, and Vietnam. Mainly found in UP, Assam and Arunachal Pradesh and the terai regions of Nepal.
- IUCN: Critically Endangered | CITES: Appendix I | Wildlife Protection Act: Schedule I



Species in News

- **White-Bellied Heron**
- Also called Imperial heron, is a brownish grey heron, is the second largest heron species.
- Distribution: It is found in India (Arunachal Pradesh, Assam), NE Bangladesh and Bhutan.
- Habitat: Inland swamp forests, forested rivers and submontane grasslands.
- IUCN: Critically Endangered | Wildlife Protection Act: Schedule I



Species in News

- **White-bellied blue robin**
- It is a bird of the family Muscicapidae.
- **Habitat:** It is endemic to the Shola forests of the higher hills of southern India. (found on the forest floor and undergrowth of dense forest patches in Shola forests)
- **IUCN Status:** Vulnerable
- **Jerdon's Courser**
- Jerdon's courser is a nocturnal bird considered to be extinct until its rediscovery in 1986.
- **Distribution:** It is endemic to Eastern Ghats (found only in Andhra Pradesh and Telangana).
- **Habitat:** Scrub jungles of the Sri Lankamaleswara WLS
- **IUCN Status:** CR | **WPA:** Sch I



Species in News

- **Manipur Bush Quail**
- Distribution: North-eastern India and Bangladesh
- Threats- habitat is small, fragmented, and rapidly shrinking
- IUCN: Endangered | Wildlife Protection Act: Schedule I



Species in News

- **Ludlow's Bhutan Glory**
- Habitat- endemic to Arunachal Pradesh, Bhutan
- Threat- unregulated grazing, forest fires, and high consumption of wood, conversion of habitat to arable land, climate change
- IUCN Status- Endangered
- **Forest Owlet**
- Habitat: South Madhya Pradesh, in north-west Maharashtra and north-central Maharashtra (endemic to the forests of central India)
- Threats- Logging operations, burning and cutting of trees for firewood, Encroachment for cultivation, grazing
- IUCN: Endangered | CITES: Appendix I | Wildlife Protection Act: Schedule I



- **Ganges Shark**
- It is a freshwater shark found only in rivers and possibly estuaries.
- Its small eyes (like dugong) suggest it is adapted to turbid water (cloudy water with poor visibility).
- Distribution: It is endemic to India and Bangladesh. Its major population is found in Ganges and Brahmaputra River Systems (India) and Padma River (Bangladesh).
- It is also found in Mahanadi River (India),
- Indus River System (NW India & Pakistan), Yangon River (Myanmar), Malayasia and Indonesia.

Species in News



- Habitat: Tropical and sub-tropical rivers and estuaries. It can survive in muddy waters.
- Threats: Over fishing, habitat degradation, building of dams and pollution
- IUCN Status- Critically Endangered | Wildlife Protection Act- Schedule I



Species in News

- **Orange Finned Mahseer or Hump Backed Mahseer**
- It is a large freshwater fish. It is the largest mahseer species.
- It is known as Tiger of Water or Tiger of Cauvery due to the fight it puts up during fishing.
- **Distribution:** It is endemic to the Cauvery River basin in the Western Ghats.
- It is an important bio-indicator for the health of the Cauvery river.
- **Habitat:** Inland freshwater and wetlands.
- **Threats:** Construction of dams, destructive fishing methods, reduced water flow rates, pollution, etc
- **IUCN Status-** Critically Endangered | **CITES-** Appendix I | **Wildlife Protection Act-** Schedule I



Species in News

- **Golden Mahseer**
- Golden mahseer (or Himalayan mahseer) or is a large golden colour fish endemic to Himalayan region.
- It is toughest among the fresh water sport fish and so, it is called Tiger of Indian Rivers.
- It is also the state fish of the states of Arunachal Pradesh, HP, Uttarakhand, MP and the UT of J&K.
- Distribution: Basins of the Indus, Ganges and Brahmaputra rivers.
- Habitat: High energy streams and rivers in montane and submontane region. Also, adapted itself to thrive in dams and lakes.
- Threats: Habitat degradation, fishing, recreational activities, pollution and natural disasters
- IUCN Status- Endangered



- **Blue Finned Mahseer**
- It is a freshwater fish. It is also known as Deccan Mahseer or Tor Khudree.
- Features: It is a silvery-bluish coloured fish with blood-red fins or fins tipped with a bluish tinge.
- Indicator species: Blue finned Mahseer is very sensitive to dissolved oxygen levels, water temperature and sudden climatic changes. It just cannot bear pollution.
- Habitat: Blue Finned Mahseer is found in the Mula-Mutha River close to the Indian city of Pune, a part of the Krishna River basin. It is also found in other rivers of the Deccan Plateau.
- IUCN Status: Least Concern (LC)

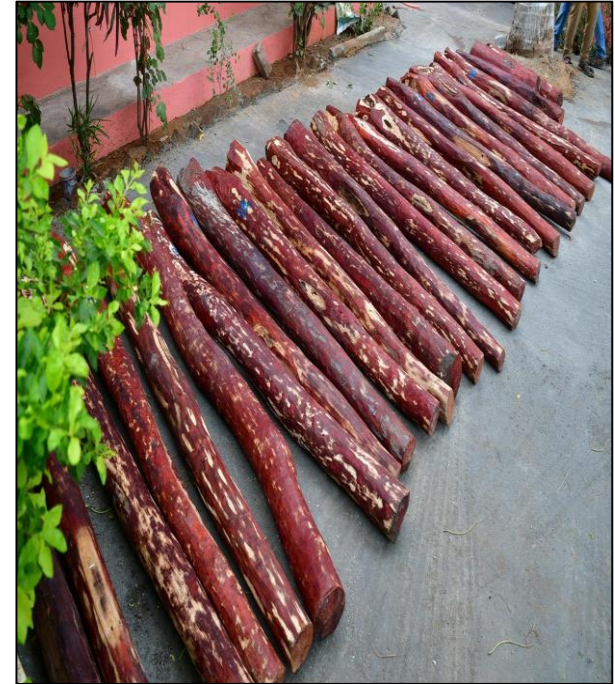


- **Large-tooth Sawfish**
- **Distribution:** Western part of the Indo-Pacific (East Africa to New Guinea, Philippines and Vietnam to Australia). In India, it is known to enter the Mahanadi river, up to 64 km inland, and also is very common in the estuaries of the Ganga and Brahmaputra.
- **Threats:** The principal threat to all sawfish are fisheries. Their long tooth-studded saw makes them extraordinarily vulnerable to entanglement in any sort of netgear.
- **IUCN Status: CR**



Species in News

- **Red Sanders or Pterocarpus santalinus**
- Characteristics:
- Used in cosmetics, medicinal products and high-end furniture/woodcraft.
- Well-drained red soils with graveled loam are suitable for cultivation.
- Regenerates well in a dry hot climate and requires rainfall ranging from 800 mm to 1000 mm annually for excellent growth.
- Habitat: Endemic to a distinct tract of forests in Eastern Ghats region of Andhra Pradesh. Mainly found in the Seshachalam forests and Palakonda hills of Chittoor, YSR (Kadapa), Annamayya and Tirupati districts of AP)
- IUCN Red List: Endangered | CITES: Appendix II



Species in News

- **Neelakurinji**
- **Context**
- Added to Schedule III of Wildlife Protection Act
- Grows at an altitude of 1,300 to 2,400 meters.
- Acts like a soil binder in grasslands and is fire hardy and resistant.
- Blooms once every 12 years season.
- Habitat: It is endemic to Western ghats generally bloom in the regions of Tamil Nadu, Kerala & Karnataka.



Species in News

- **Gucchi Mushroom**
- Is a species of fungus in the family Morchellaceae of the Ascomycota
- They are pale yellow in colour with large pits and ridges on the surface of the cap, raised on a large white stem.
- It is primarily obtained from the forests and pastures of Doda district and it is also found in the high altitude areas of Kupwara, Pahalgam, Shopian, Kishtwar and Poonch of Jammu & Kashmir.
- It is one of the costliest mushrooms in the world which is known for its spongy, honeycombed head and savoury flavour.
- Benefits- They are rich in antioxidant and antimicrobial properties that prevent health issues including heart diseases and diabetes by removing reactive oxygen species that harm the body



- **Context**
- **Three Himalayan medicinal plant species have made it to IUCN Red List.**
- **Meizotropis pellita (IUCN: Critically Endangered)**
- Endemic to Uttarakhand.
- Oil extracted from leaves possesses strong antioxidants and can be natural substitute for synthetic antioxidants in pharmaceutical industries.
- **Fritillaria cirrhosa (IUCN: Vulnerable)**
- Used in treatment of bronchial disorders and pneumonia.
- **Dactylorhiza hatagirea (IUCN: Endangered)**
- Used in Ayurveda, Siddha, Unani, and other alternative systems of medicine to cure fever, cough etc.

State Symbols



- **Reasons for declaration**
- Since certain animals are endemic to certain parts of the country, declaring them as state emblems garnered special attention for them.
- It also gives the people of the state a sense of pride in that species, a part of their natural heritage
- Eg- In 1975, when the Sangai or brow-antlered deer of Manipur was declared the state animal, there were only 14 individuals left. However, following the declaration, people started to own the Sangai. Today, it is an icon of Manipur.

State Symbols



S.No.	States	Animal	Bird	Tree	Flower	Butterfly
1.	Andhra Pradesh	Blackbuck	Indian Roller	Neem	Water lily	*
2.	Arunachal Pradesh	Mithun	Great Hornbill	Hollong	Foxtail Orchid	*
3.	Assam	Indian One-horned Rhino	White-winged Wood Duck	Hollong	Foxtail Orchids	*
4.	Bihar	Gaur	Indian Roller	Peepal	Kachnar	*
5.	Chhattisgarh	Wild Buffalo	Hill Myna	Sal	Indian Laurel	*
6.	Delhi	Nilgai	House sparrow	Gulmohar	Alfalafa	*
7.	Goa	Gaur	Black Crested Bulbul	Matti	Plumeria Rubra	*
8.	Gujarat	Asiatic Lion	Greater Flamingo	Banyan	Marigold (Galgota)	*
9.	Haryana	Blackbuck	Black Francolin	Peepal	Lotus	*
10.	Himachal Pradesh	Snow Leopard	Western Tragopan	Deodar	Rhododendron	*
11.	Jammu Kashmir &	Hangul	Black necked Crane	Chinar	Rhododendron	*
12.	Jharkhand	Elephant	koel	Sal	Palash/Parrot Tree	*
13.	Karnataka	Elephant	Indian Roller	Sandal wood	Lotus	Southern Birdwing
14.	Kerala	Elephant	Great Hornbill	Coconut	Kanikonna/Golden Shower Tree	*
15.	Madhya Pradesh	Swamp Deer	Paradise Fly catcher	Banyan	Palash/Parrot Tree	*

State Symbols



16.	Maharashtra	Giant Squirrel	Green Imperial Pigeon	Mango	Jarul	Blue Mormon
17.	Manipur	Sangai	Mrs. Hume's Pheasant	Toon	Siroi Lily	*
18.	Meghalaya	Clouded Leopard	Hill Myna	Gamari	Lady Slipper Orchid	*
19.	Mizoram	Serow	Mrs. Hume's Pheasant	Iron Wood	Red Vanda	*
20.	Nagaland	Mithun	Blyth's Tragopan	Alder	Rhododendron	*
21.	Orissa	Elephant	Peacock	Banyan	Ashoka	*
22.	Punjab	Blackbuck	Northern Goshawk	Sheesham	Gladiolus	*
23.	Rajasthan	Chinkara	Indian Bustard	Khejri	Rohira	*
24.	Sikkim	Red Panda	Blood Pheasant	Rhododendron	Noble Orchid	*
25.	Tamil Nadu	Nilgiri Tahr	Emerald Dove	Palmyra Palm	Kandhal	*
26.	Telangana	Spotted Deer (Jinka)	Indian Roller (Paalapitta)	Shami (Jammi chettu)	Tanner's Cassia	*
27.	Tripura	Phayre's Langur	Green Imperial Pigeon	Agar	Nageshwar	*
28.	Uttarakhand	Musk Deer	Himalayan Monal	Burans	Brahm Kamal	Common Peacock
29.	Uttar Pradesh	Swamp Deer	Sarus Crane	Ashok	Brahm Kamal	*
30.	West Bengal	Fishing cat	White-throated King Fisher	Devil's Tree	Shephali	*

State Symbols



S.No.	Union Territories	Animal	Bird	Tree	Flower	Butterfly
31.	Andaman & Nicobar Islands	Dugong	Andaman Wood Pigeon	Andaman Padauk	Pyinma Tree (Lagerstroemia hypoleuca)	*
32.	Chandigarh	Indian grey Mongoose	Indian grey Hornbill	Mango	Dhak	*
33.	Dadar & Nagar Haveli	*	*	*	*	*
34.	Daman & Diu	*	*	*	*	*
35.	Lakshwadeep	Butterfly Fish	Sooty Tern	Bread Fruit	Neelakurinji	*
36.	Puducherry	Squirrel	Asian Koel	Bael Fruit Tree	Cannon ball flower (Nagalinga flower)	*

National Conservation Efforts



- **Integrated Development of Wildlife Habitats**
- Integrated Development of Wildlife Habitats (IDWH) is an on-going Centrally Sponsored Scheme.
- Under IDWH, financial assistance is provided to State/UTs for protection and conservation of wildlife.
- Components of the Scheme
 1. Support to Protected Areas (National Parks, Wildlife Sanctuaries, Conservation and Community Reserves)
 2. Protection of Wildlife Outside Protected Areas
 3. Species Recovery Programmes for saving Indian critically endangered species and habitats.

National Conservation Efforts



- **22 species are included in the programme**
- Mammals
 - 1. Hangul (Kashmir Stag – CR)
 - 2. Malabar Civet (CR)
 - 3. Asian Wild Water Buffalo (EN)
 - 4. Asiatic Lion (EN)
 - 5. Brow-Antlered Deer / Sangai / Eld's Deer (EN)
 - 6. Nilgiri Tahr (EN)
 - 7. Red Panda (EN)

National Conservation Efforts



- 8. Clouded Leopard (VU)
- 9. Indian Rhino / Great One-Horned Rhinoceros (VU)
- 10. Snow Leopard (VU)
- 11. Swamp Deer / Barasingha (VU)
- 12. Caracal Cat (LC) (most recent addition to the list)
- **Marine Mammals**
- 1. Dugong (VU)
- 2. Gangetic River Dolphin (EN)

National Conservation Efforts



- **Birds**

- 1. Great Indian Bustard (CR)
- 2. Jerdon's Courser (CR)
- 3. Nicobar Megapode (VU)
- 4. Edible Nest Swiftlet (LC)
- 5. Vultures (all species found in India)

- **Reptiles**

- 1. Arabian Sea Humpback Whale (EN) (it is a subpopulation of Humpback Whales (LC))
- 2. Marine Turtles
- 3. Northern River Terrapin (CR)

National Conservation Efforts



- **National Wildlife Action Plan (NWAP) 2017-2031**
- 1st NWAP was adopted in 1983.
- 2nd NWAP was put in place for the period (2002-2016).
- 3rd NWAP is for the period 2017-31.
- It has 5 components & 103 conservation actions.
- NWAP adopts the landscape approach of conservation by emphasizing on preservation of genetic diversity and sustainable utilization of species and ecosystem.
- Landscape Approach to Conservation
- It is a holistic approach to landscape management, aiming to reconcile the competing objectives of nature conservation and economic activities across a given landscape.

Invasive Alien Species



- **Definition**
- An organism that causes ecological or economic harm in a new environment where it is not native.
- For E.g.: In India, Parthenium (came from wheat imported from U.S. in 1950s) and lantana (brought by British as ornamental plants from South America) threatens more than 40% of India's tiger reserves.
- **Characteristics:**
- Multiply quickly due to no competitors or predators
- High dispersal rate
- They have a wide geographic range, abundant, easily dispersed seed production, robust root growth, and quick generation times

Invasive Alien Species



- **Examples**

- **Juliflora Prosopis**

- It is native to Mexico, South America and the Caribbean.
- Tamil Nadu- Seemai karuvelam- was used as for firewood initially but got over-spread, 2 years back Madras High Court asked the Government to weed out this plant.

- **House Pigeons**

- Proliferation of blue rock pigeons to the detriment of other once common such as the house sparrow and the Brahminy myna

Invasive Alien Species



- **Tilapia**
- They tolerate a wide range of conditions — from Chennai's toxic Cooum river to the pristine waters of many of India's national parks and sanctuaries.
- The species were introduced globally from the remote islands of the South Pacific through Southeast Asia and the Americas.
- Brought for aquaculture initially
- **Burmese python**
- Native to Southeast Asia. However, since the end of the 20th century, they have become an established breeding population in South Florida
- Now classified as invasive specie in Florida
- Irula tribesmen from Tamil Nadu are helping in capturing the snakes

Invasive Alien Species



- **Senna spectabilis**
 - Has taken over between 800 and 1,200 hectares of the buffer zones of the Mudumalai Tiger Reserve (MTR)
 - It is an exotic tree with bright yellow flowers.
 - Introduced as an ornamental species and for use as firewood from South and Central America.
 - Negative effect on local biodiversity, crowding out native species and limiting food availability for wildlife.
- **Red-eared slider turtle**
 - Mature fast, grow larger, and produce more offspring, thus, threat to native freshwater turtle.
 - Can out-compete native turtles for food, nestling, and basking sites.
 - Can transfer diseases and parasites to native reptile species.

2013

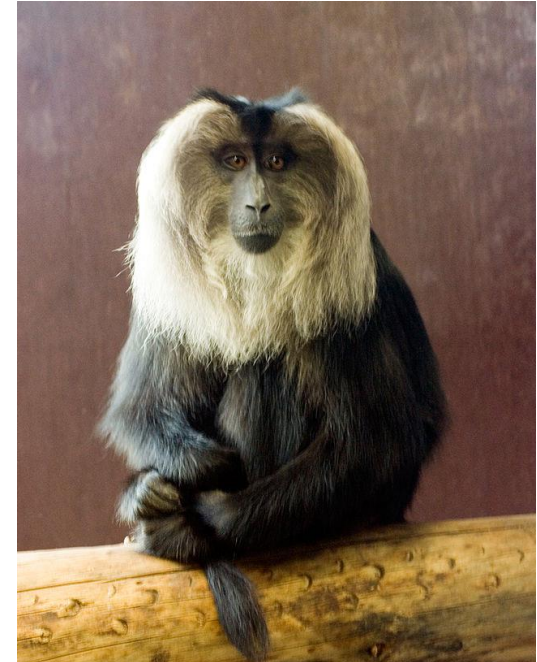
In which of the following States is lion-tailed macaque found in its natural habitat?

1. Tamil Nadu
2. Kerala
3. Karnataka
4. Andhra Pradesh

Select the correct answer using the codes given below.

- a) 1, 2 and 3 only
- b) 2 only
- c) 1, 3 and 4 only
- d) 1, 2, 3 and 4

- **Answer: A (1, 2 and 3 only)**
- The lion-tailed macaque (*Macaca silenus*), or the wanderoo, is an Old World monkey endemic to the Western Ghats of South India.
- A recent assessment for IUCN reports 3000-3500 of these animals live scattered over several areas in Tamil Nadu, Kerala, and Karnataka.



2013

Consider the following fauna of India:

1. Gharial
2. Leatherback turtle
3. Swamp deer

Which of the above is/are endangered?

- a) 1 and 2 only
- b) 3 only
- c) 1, 2 and 3 only
- d) None

- **Answer: C**
- The gharial (*Gavialis gangeticus*), also known as the gavial, and the fish-eating crocodile is a crocodilian in the family Gavialidae and is native to the northern part of the Indian subcontinent. As the population has declined drastically since the 1930s, the gharial is listed as Critically Endangered on the IUCN Red List.



- The leatherback sea turtle (*Dermochelys coriacea*), sometimes called the lute turtle or leathery turtle or simply the luth is the largest of all living turtles and is the fourth-heaviest modern reptile behind three crocodilians. It is listed as **Vulnerable** on the IUCN red list
- The barasingha also called swamp deer, is a deer species distributed in the Indian subcontinent. Populations in northern and central India are fragmented, and two isolated populations occur in southwestern Nepal. It is listed as **Vulnerable** on the IUCN red list



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- The barasingha also called swamp deer, is a deer species distributed in the Indian subcontinent. Populations in northern and central India are fragmented, and two isolated populations occur in southwestern Nepal. It is listed as **Vulnerable** on the IUCN red list



2014

Consider the following animals:

1. Sea cow
2. Sea horse
3. Sea lion

Which of the above is/are mammal/mammals?

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1, 2 and 3

- **Answer: B**
- Sea cow, (*Hydrodamalis gigas*), also called Steller's sea cow, a very large aquatic mammal, is now extinct. It belonged to the family Dugongidae and Dugong.
- Sea Horse belongs to the class Actinoptery and genus Hippocampus of Pisces (fish). Its head and neck resemble that of a horse and are found in seas. It is a bony fish (Osteichthyes).
- Sea lions belong to the family Otariidae of class Mammalia. These are carnivorous aquatic mammals. It is the largest of all sea lions.

2013

Consider the following

1. Star tortoise
2. Monitor lizard
3. Pygmy Hog
4. Spider monkey

Which of the above are found in India?

- a) 1, 2 and 3 only
- b) 2 and 3 only
- c) 1 and 4 only
- d) 1, 2, 3 and 4

Answer: A

- Indian Star Tortoise is found in India, Sri Lanka, and Pakistan. It is an endangered species as per IUCN Red List.
- Monitor Lizards are native to Asia, Africa, and America. There are four kinds of Monitor Lizards in India.
- Pygmy Hog is a scientific name of Porcula Sylvania. It is considered endangered on the IUCN Red List. It is protected in the Manas National Park of Assam.
- Spider Monkey belongs to the genus Ateles and is found in the tropical forests of Central and South America. **Hence, 4 is incorrect.**

2014

Other than poaching, what are the possible reasons for the decline in the population of Ganges River Dolphins?

1. Construction of dams and barrages on rivers
2. Increase in the population of crocodiles in rivers
3. Getting trapped in fishing nets accidentally
4. Use of synthetic fertilizers and other agricultural chemicals in crop fields in the vicinity of rivers

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1, 3 and 4 only
- d) 1, 2, 3 and 4

Answer:

C

The Ganges river dolphin was officially discovered in 1801. **Ganges river dolphins** once lived in the **Ganges-Brahmaputra-Meghna and Karnaphuli-Sangu river systems of Nepal, India, and Bangladesh**. But the species is extinct from most of its early distribution ranges.

- The Ganges river dolphin can only live in freshwater and is essentially blind.

- Reasons for declining population-
 - Habitat loss and degradation
 - Pollution
 - Fisheries bycatch- Bycatch in gillnets and line hooks is also a major source of mortality for this subspecies.
 - Directed take

2015

With reference to 'dugong', mammal found in India, which of the following statements is/are correct?

1. It is a herbivorous marine animal.
2. It is found along the entire coast of India
3. It is given legal protection under Schedule 1 of the Wildlife (Protection) Act, 1972.

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 3 only

Answer: C

- The dugong is a medium-sized marine mammal.
- It is one of four living species of the order Sirenia, which also includes three species of manatees.
- It is the only living representative of the once-diverse family Dugongidae; its closest modern relative, Steller's sea cow (*Hydrodamalis gigas*), was hunted to extinction in the 18th century. The dugong is the only strictly herbivorous marine mammal.
- As dugongs had been brought under Schedule I of the Wildlife Protection Act, the punishment for poaching is imprisonment, but seldom is anyone punished.
- The offenders would get the same punishment as that of killing tigers, leopards or elephants.
- It is not found along the entire coast of India

2015

Which one of the following is the national aquatic animal of India?

- a) Saltwater crocodile
- b) Olive ridley turtle
- c) Gangetic dolphin
- d) Gharial

Answer: B

The National Aquatic animal of India is the Ganges River Dolphin (*Platanista gangetica*).

- The dolphin is blind and faces a severe threat from the heavy pollution and habitat destruction in the waters where it lives.
- It has been declared Endangered in the IUCN red list and it is assumed that less than 2000 individuals remain.
- This dolphin, called 'susu' or 'susuk' in India, is thought to represent the purity of The Ganga.

2016

In which of the following regions of India are you most likely to come across the 'Great Indian Hornbill' in its natural habitat?

- a) Sand deserts of northwest India
- b) Higher Himalayas of Jammu and Kashmir
- c) Salt marshes of western Gujarat
- d) Western Ghats

Answer: D

Great hornbills are found in the **forests of India, Bhutan, Nepal, Mainland Southeast Asia, Indonesian Island of Sumatra and Northeastern region of India.**

- The distribution of the species is fragmented over its range in the Indian subcontinent and Southeast Asia.
- In the subcontinent, they are found in a few forest areas in the **Western Ghats and in the forests along the Himalayas.**

2016

Q.15) Recently, for the first time in our country, which of the following States has declared a particular butterfly as 'State Butterfly'?

- a) Arunachal Pradesh
- b) Himachal Pradesh
- c) Karnataka
- d) Maharashtra

Answer: D

- Maharashtra has become the first State in the country to have a 'State butterfly.'
- The state government has declared the Blue Mormon (*Papilio polymnestor*) as the State butterfly.
- The decision was taken at a meeting of the State Wildlife Board in Mumbai
- The Blue Mormon is a large, swallowtail butterfly found primarily in Sri Lanka and India, mainly restricted to the Western Ghats of Maharashtra, South India and coastal belts.
- It may occasionally be spotted in the Maharashtrian mainland between Vidarbha and Western Maharashtra.

2017

The term 'M-STRIPES' is sometimes seen in the news in the context of

- a) Captive breeding of Wild Fauna
- b) Maintenance of Tiger Reserves
- c) Indigenous Satellite Navigation System
- d) Security of National Highways

Answer: B

- Monitoring System for Tigers'- Intensive Protection and Ecological Status is a software monitoring system launched by the Indian Government in 2010 in some tiger reserves to **reduce the vulnerability of Tigers**.
- The system would enable field managers to assist intensity and spatial coverage of patrols in a geographic information system (GIS) domain. **Hence, option b is correct.**

2017

Recently there was a proposal to translocate some of the lions from their natural habitat in Gujarat to which one of the following sites?

- a) Corbett National Park
- b) Sariska National Park
- c) Mudumalai Wildlife Sanctuary
- d) Kuno Palpur Wildlife Sanctuary

Answer: D

- An environment ministry's expert committee approved Kuno Palpur in Madhya Pradesh as the second home for Asiatic lions found only in Gir national park. However, the Gujarat government will not share lions unless 33 studies as mandated by the international wildlife watchdog IUCN are completed. **Hence, option d is correct.**

2019

Q.10) Consider the following pairs:

Wildlife : Naturally found in

1. Blue-finned Mahseer: Cauvery River
2. Irrawaddy Dolphin: Chambal River
3. Rusty-spotted Cat: Eastern Ghats

Which of the pairs given above are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer: C

- Blue-finned Mahseer is found naturally in the Cauvery river. **Hence, pair 1 is correct.**
- The Irrawaddy dolphin is a critically endangered species. It is found near sea coasts and in estuaries and rivers in parts of the Bay of Bengal and Southeast Asia. In India, it is found in Lake Chilika and not the Chambal river. The Gangetic dolphin is found in the Chambal river. **Hence, pair 2 is incorrect.**

Answer: C

- Blue-finned Mahseer is found naturally in the Cauvery river. **Hence, pair 1 is correct.**
- The Irrawaddy dolphin is a critically endangered species. It is found near sea coasts and in estuaries and rivers in parts of the Bay of Bengal and Southeast Asia. In India, it is found in Lake Chilika and not the Chambal river. The Gangetic dolphin is found in the Chambal river. **Hence, pair 2 is incorrect.**

- Rusty Spotted Cat is one of the world's smallest feline. The rusty-spotted cat, one of the few wild cats that inhabit the forests of Andhra Pradesh, is among the animals in the Eastern Ghats. **Hence, pair 3 is correct.**

2020

Which of the following are the most likely places to find the musk deer in its natural habitat?

1. Askot Wildlife Sanctuary
2. Gangotri National Park
3. Kishanpur Wildlife Sanctuary
4. Manas National Park

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 3 and 4 only
- (d) 1 and 4 only

Solution

(A) 1 and 2 only

- Musk deer is a small compact deer belonging to the family Cervidae
- Askot Musk Deer Sanctuary has located 54 km from Pithoragarh near the town of Askot in Uttarakhand.
- Gangotri National Park is a national park in the Uttarkashi District of Uttarakhand in India.
- While Kishanpur WS is located in UP and Manas NP in Assam

2020

With reference to Indian elephants, consider the following statements:

1. The leader of an elephant group is a female.
2. The maximum gestation period can be 22 months.
3. An elephant can normally go on calving till the age of 40 years only.
4. Among the States in India, the highest elephant population is in Kerala.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 4 only
- c) 3 only
- d) 1, 3 and 4 only

Solution

(A) 1 and 2 only

- Karnataka leads the table with 6,049 elephants followed by Kerala
- Elephants are led by a matriarch and have gestation period of 22 months.

2022

Which of the following is not a bird?

- a) Golden Mahseer
- b) Indian Nightjar
- c) Spoonbill
- d) White Ibis

Answer: A

- Golden mahseer (or Himalayan mahseer) or is a **large golden colour fish** endemic to Himalayan region.
- It is toughest among the fresh water sport fish and so, it is called Tiger of Indian Rivers.
- It is also the state fish of the states of Arunachal Pradesh, HP, Uttarakhand, MP and the UT of J&K.



2016

With reference to 'Red Sanders', sometimes seen in the news, consider the following statements:

1. It is a tree species found in a part of South India.
2. It is one of the most important trees in the tropical rainforest areas of South India.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Answer: A

- Red sanders are one of the most important trees in the tropical deciduous areas of South India. So, the answer is a) 1 only

2021

Consider the following animals:

1. Hedgehog
2. Marmot
3. Pangolin

to reduce the chance of being captured by predators, which of the above organisms rolls up/roll up and protects/protect its/their vulnerable parts?

- a) 1 and 2
- b) 2 only
- c) 3 only
- d) 1 and 3

Answer: D

- Hedgehogs are a small mammal with short limbs and a body low to the ground. When they are frightened, or annoyed, hedgehogs will roll into a ball so that a predator will feel the full brunt of its sharp spines and will then leave the hedgehog alone. **Hence option 1 is correct**



- Marmots are relatively large ground squirrels in the genus *Marmota*, with 15 species living in Asia, Europe, and North America. These herbivores are active during the summer when often found in groups, but are not seen during the winter when they hibernate underground. They are the heaviest members of the squirrel family. They do not roll up when threatened. **Hence, option 2 is not correct.**
- Pangolins are uniquely covered in tough, overlapping scales. These mammals eat ants and termites using an extraordinarily long, sticky tongue, and are able to quickly roll themselves up into a tight ball when they feel threatened. **Hence option 3 is correct.**